

Paying to Remember: Price Geometry and Renewal Control of LLM Context Compaction

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Abstract

Long-running language-model agents repeatedly pay to present their working history. Compaction shortens that history but also generates a summary, rewrites cacheable prefixes, and restores documents needed to continue work. Consequently, fewer context tokens need not imply a lower invoice. We develop a stochastic account of this trade-off that separates input, cache-write, cache-read, and output prices. For regenerative workloads, integrated renewal occupation yields a distribution-general control rule balancing reconstruction expense against repeated history carrying. The rule applies to both continuous growth laws and integer-token plateaus. Its comparative statics identify a cache-reliability boundary at which increasing the write price reverses the preferred compaction direction. A cold crossing-tail correction exposes a second reversal mechanism, while price-envelope geometry separates policy response from invoice response and explains why retuning losses are often small. We instantiate the analysis with 297,091 public coding-agent growth observations, matched independent and block-resampled workloads, and 38 quoted price vectors. Under a common workload, a very low read price supports a 162k-token threshold at perfect hits but only 94k at an 80% hit probability. Reclassifying existing material as stable and adding stable material move the threshold in opposite directions. Local correlation increases invoice variance much more than it changes the long-run minimum; short remaining horizons favor delaying resets. The resulting perspective treats compaction as amortized state reconstruction, placing cache reliability and working-state design ahead of microscopic threshold tuning.

Keywords

LLM agents, context compaction, prompt caching, renewal processes, sensitivity analysis, workload characterization

1 Introduction

A coding agent reads an implementation, inspects a failing test, edits a file, and runs another command. Each interaction adds evidence to its working history. That history is useful twice: semantically, because it explains the current state of the task, and computationally, because an unchanged prefix can reuse cached inference state. Eventually the agent compacts its context. A short summary replaces many earlier interactions, but the repository still has to be understood. The next edit may require the agent to see a file again, and the rewritten transcript may have to be cached anew. What looked like the removal of expensive history has become the purchase of a new working state.

This tension is becoming more consequential as agent sessions grow longer. Million-token windows make continued accumulation technically possible, while tool-driven trajectories create bursts of logs, source files, and observations that are expensive to retain

indiscriminately. Production context-management systems now combine summaries, recent turns, and refreshed files rather than replacing the entire state with a tiny paragraph [1, 22, 23]. Public coding-agent traces expose repeated prefixes, tool-mediated growth, and human-paced gaps [24]. At the same time, API tariffs increasingly distinguish ordinary input, cache creation, cache reuse, and generated output [3, 7, 14, 18]. Context length and invoice cost have therefore ceased to be interchangeable quantities.

Consider the apparently simple instruction to compact earlier when cache writes become more expensive. With unreliable reuse, earlier compaction can avoid repeatedly rewriting a growing history. With reliable reuse, however, that history is mostly read at a discount, whereas compaction forces the volatile working state to be written again. More expensive writes can then make *later* compaction preferable. Even this distinction is incomplete: the latest generated tail may remain cold when the compactor consumes it, so compaction can avoid a write that a later ordinary request would otherwise pay. The policy direction depends on the interaction between prices, eligible prefixes, reconstruction, and the stopping time of the next reset.

We ask: *what structure determines the invoice-optimal compaction line, and which changes materially alter that decision?* Our focus is the economics of a declared continuation workload. We hold ordinary work fixed, account for the paid housekeeping it induces, and vary reconstruction, cache, and price mechanisms explicitly. This isolates the decision about when to rebuild from the separate problem of choosing a compression representation that preserves task utility. It also provides a foundation on which utility and adaptive memory decisions can subsequently be layered.

The central observation is that compaction buys a cycle of cheap continued work. If S is the reconstructed context and L the additional growth allowed before the next reset, the relevant quantity is neither L alone nor the average increment. It is the *integrated renewal occupation* $D(L)$, which accounts for how many requests visit each history size before crossing the trigger. A reconstruction-to-carry price ratio determines the crossing $D(L_*) = A/c$. This relation preserves stochastic overshoot and works across growth distributions, including finite integer-valued observations.

We organize the contribution around three findings.

- **Compaction has a price-conditioned phase structure.** The renewal crossing yields explicit price-share sensitivities and a write-price reversal boundary. Cold terminal content supplies an additional, independently identifiable mechanism for reversing the core prediction.
- **Policy sensitivity and invoice sensitivity obey different laws.** Relative prices govern the threshold, whereas optimized invoice derivatives are expected resource use.

A scalar smooth control has rank-one price curvature, explaining how visible threshold movement can produce little economic benefit from retuning.

- **Workload evidence identifies where the structure matters.** Community-derived increments and matched dependence controls show that cache reliability, reconstruction composition, and remaining work can be more important than selecting a microscopic argmin. A broad tariff study reveals shared price-ratio classes and cache-mode trade-offs.

The practical shift is from asking how much text a compressor removes to asking what state it reconstructs, what history remains cheap to reuse, and how much future work can amortize the reset.

2 Related Work

Prompt compression and agent memory. Prompt-compression research has developed techniques for preserving salient information with fewer tokens. LLMingua uses a coarse-to-fine compression pipeline, while LLMingua-2 learns task-agnostic extractive compression from distilled data [8, 16]. These methods primarily address the representation of a prompt. In an agent, the same representation is presented repeatedly and can be rewritten repeatedly. Recent studies compare observation masking, truncation, and generated summarization in that setting. Lindenbauer et al. find that simple observation masking can compete with summarization [9]; Satish et al. separate compression primitives and their effects on agent workloads [17]. Together, this literature motivates moving beyond a one-shot token-reduction objective. Our analysis takes the continuation state as the unit of accounting and derives how repeated carry and reconstruction determine the *timing* of compression.

Cache-compatible context management. Prefix caching turns unchanged context into a reusable computational asset. Its value depends on byte-level prefix structure, lookup boundaries, retention, and routing, rather than semantic similarity alone. Provider documentation makes these distinctions operational [2, 6, 13, 18]. Lumer et al. evaluate caching strategies for long-horizon agentic tasks [10]; TokenPilot connects cache-efficient segmentation with recovery of externally stored payloads [22]. This work places cache layout inside the context-management problem. We complement that design perspective with a four-price stochastic ledger: cache creation and reuse enter different sides of the amortization balance, and their interaction produces explicit policy-direction reversals. The cold crossing tail further links cache eligibility to the point at which a reset is triggered.

Working sets and durable state. The distinction between resident context and recoverable information echoes Denning’s working-set model [4]. MemGPT applies hierarchical memory and explicit transfers to language-model agents [15]; SWE-agent’s agent–computer interface retains repository state while controlling displayed observations [23]. These systems suggest that forgetting prompt text need not destroy the underlying artifact, yet the artifact must be made available when the next operation needs it. We study the concentrated reconstruction regime: a project phase repeatedly returns to a similar aggregate working state. This regime makes the

price mechanisms analytically visible. First-use retrieval, version-dependent prerequisites, and phase changes lead naturally to richer state-dependent controllers, discussed in Section 5.

Workload characterization and comparative statics. TraceLab provides public request-level coding-agent traces with which to study growth, caching, and serving behavior [24]. Our empirical law is estimated from chronological full-input changes and retains large tool-mediated jumps. A matched block design distinguishes short-range dependence from changes in the one-step workload distribution. Classical checkpoint/restart analysis likewise balances recurring reset expense against losses accumulated between resets [5]. Its square-root intuition is a useful starting point. Here the exposure is paid history carrying rather than failure-driven lost computation, and reconstruction can preserve one cached prefix while rewriting another. Stochastic crossing and a cold terminal tail determine how that balance is realized. Renewal-reward arguments expose the endogenous cycle length, while envelope theory characterizes optimized price response [12]. The value of combining these perspectives is a mechanism-based sensitivity map: a result can predict the direction of a change before a simulation measures its size, and a simulation can reveal when a neglected tail or finite horizon changes the prediction.

3 Model and Theory

3.1 A cycle of continued work and reconstruction

We measure work by ordinary agent actions. A paid compaction is housekeeping between actions; it does not complete an extra unit of ordinary work. A cycle starts with reconstructed context S . This comprises the retained prefix, summary, and restored material needed for the next project phase. Let B be the eligible unchanged prefix within S , and let C be newly generated summary content, also within S . Thus $0 \leq B$, $0 \leq C$, and $B + C \leq S$. The volatile rewritten portion is $S - B$. The compactor adds K instruction tokens; each ordinary action has I fresh instruction tokens.

For ordinary action n , let $G_n \geq 0$ be retained context growth and $O_n \geq 0$ its billed output. Growth includes retained output and tool observations, so output is charged as output without being inserted into G_n a second time. In the regenerative model, the pairs (G_n, O_n) are iid and independent of cache marks, with $0 < g = \mathbb{E}G_n < \infty$ and $o = \mathbb{E}O_n < \infty$. Dependence within each growth/output pair is retained; output mean o enters the fixed-action invoice baseline.

The controller compacts at the next action boundary after context reaches $H > S$. With $L = H - S$, define

$$Y_n = \sum_{k=1}^n G_k, \quad T_L = \inf\{n \geq 1 : Y_n \geq L\}. \quad (1)$$

The compactor processes the crossed state $S + Y_{T_L}$, including overshoot, rather than an artificial state exactly equal to H .

The nonnegative price vector is $\mathbf{p} = (p_i, p_w, p_r, p_o)$ in currency per token. Input, creation, and reuse charges form disjoint input categories; output is charged separately. A qualifying old prefix hits with independent boundary probability q . This is a mechanism parameter, whereas a billed cache-read token fraction is weighted by prefix lengths. Their meanings differ.

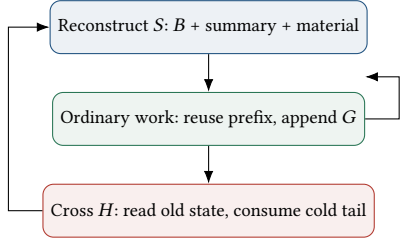


Figure 1: Compaction purchases a new working-state cycle. The stable prefix, rewritten material, and cold terminal tail have different invoice roles.

For cache timing, split an action’s growth into pre-call input $(1 - \theta)G_n$ and post-call retained tail θG_n , with $0 \leq \theta \leq 1$. The post-call tail is still cold at the next boundary. On a reset, the old eligible prefix can be read, this tail is consumed as uncached compactor input, the summary is generated, and $S - B$ is reconstructed for the next action.

3.2 Integrated occupation selects the core threshold

The renewal quantities retain the whole increment law:

$$U(L) = \sum_{n \geq 0} \mathbb{P}(Y_n < L) = \mathbb{E}T_L, \quad (2)$$

$$M(L) = \sum_{n \geq 0} \mathbb{E}[Y_n \mathbf{1}\{Y_n < L\}], \quad (3)$$

$$D(L) = LU(L) - M(L) = \sum_{n \geq 0} \mathbb{E}[(L - Y_n)_+]. \quad (4)$$

$D(L) = \int_0^L U(x) dx$ is continuous and strictly increasing even when growth is integer-valued. Repeated zero-growth actions remain in U .

Define the effective history-carry and compactor-input prices

$$c = (1 - q)p_w + qp_r, \quad \kappa = (1 - q)p_i + qp_r, \quad (5)$$

and the effective reconstruction charge

$$A = p_i[K + (1 - q)S] + qp_w(S - B) + qp_rB + p_oC. \quad (6)$$

Each coefficient has a token-accounting interpretation: old-state compactor misses, rewritten volatile material, stable compactor reads, and generated summaries. Let $\beta = p_iI + p_oO + (p_w - c\theta + \kappa)g$.

The core rate, before the cold terminal correction, is

$$j_0(L) = \beta + cS + \frac{A + cM(L)}{U(L)}. \quad (7)$$

Its unit is expected currency per ordinary action. In particular, it uses the ratio of expected cycle charge to expected cycle length, rather than the expectation of a random charge/length ratio.

THEOREM 3.1 (RENEWAL CROSSING). *For nonnegative iid growth with $0 < g < \infty$, and $A, c > 0$, the core rate is nonincreasing before the unique crossing*

$$D(L_*) = A/c \quad (8)$$

and nondecreasing after it. The crossing selects a minimizing threshold region. Where the renewal measure has density $u = U'$,

$$j'_0(L) = \frac{u(L)}{U(L)^2} [cD(L) - A]. \quad (9)$$

At a positive renewal atom, the invoice jump has the same sign.

The proof is in Appendix A. The result supplies a common control principle for smooth distributions and discrete plateaus: reconstruction is amortized until integrated history occupation reaches its relative price. For exponential growth, $U = 1 + L/g$ and $D = L + L^2/(2g)$, giving

$$L_* = \sqrt{g^2 + 2gA/c} - g. \quad (10)$$

The leading fluid approximation $\sqrt{2gA/c}$ discards the initial renewal and stochastic crossing correction. The exact empirical crossing is equally tractable through a renewal convolution.

The law also separates growth scale from dispersion. Scaling all increments by a gives $D_a(L) = aD(L/a)$ and increases the optimal gap sublinearly. At fixed mean, a convex-order increase in dispersion raises D and decreases the core gap. A variance statistic alone does not specify that distributional ordering. These mechanisms are derived in Appendix B.

3.3 Prices induce direction-changing sensitivities

Write $A = \mathbf{p}^T \mathbf{a}$ and $c = \mathbf{p}^T \mathbf{v}$, where

$$\begin{aligned} \mathbf{a} &= (K + (1 - q)S, q(S - B), qB, C), \\ \mathbf{v} &= (0, 1 - q, q, 0). \end{aligned} \quad (11)$$

Let $R = A/c$. Away from renewal atoms, differentiation of Equation (8) yields

$$\frac{\partial H_*}{\partial p_j} = \frac{a_j - Rv_j}{cU_*}. \quad (12)$$

The sign compares a price component’s reconstruction share with its carry share. The corresponding gap elasticity is

$$\epsilon_{L,p_j} = \frac{D}{LU} \left(\frac{p_j a_j}{A} - \frac{p_j v_j}{c} \right). \quad (13)$$

All four elasticities sum to zero: common scaling of token prices leaves the controller unchanged. Absolute-threshold elasticities are smaller by L/H .

PROPOSITION 3.2 (WRITE-PRICE REVERSAL). *The core write-price response has the sign of*

$$\begin{aligned} N_w(q) &= qp_r(qS - B) \\ &\quad - (1 - q)\{p_i[K + (1 - q)S] + p_oC\}. \end{aligned} \quad (14)$$

For $p_r > 0$, $S > B$, and $p_i(K + S) + p_oC > 0$, there is a unique $q_w \in [B/S, 1)$ at which this response changes from negative to positive. The boundary is independent of the growth-law family and of p_w itself.

At low q , histories repeatedly incur writes and a higher write price favors earlier compacting. At high q , histories are mainly read and a higher write price favors postponing the rewritten reset. Input and summary-output prices increase the core gap through their setup coefficients. Read prices generally decrease it; a sufficiently large stable prefix and low hit probability can instead make

recurring compactor reads dominate. Appendix C gives the precise boundary and mixed derivatives.

Material interventions must also identify their path. Reclassifying existing tokens as stable holds S fixed, giving

$$\left. \frac{\partial H_*}{\partial B} \right|_S = -\frac{q(p_w - p_r)}{cU_*}. \quad (15)$$

Adding stable material increases S and B together; adding summary while preserving other material increases S and C together. Their responses are

$$\frac{dH_*}{dB_{\text{added}}} = 1 + \frac{\kappa}{cU_*}, \quad (16)$$

$$\left. \frac{dH_*}{dC} \right|_{\text{preserve}} = 1 + \frac{p_o + (1-q)p_i + qp_w}{cU_*}. \quad (17)$$

The same phrase, “more stable context,” can therefore describe opposite threshold responses.

3.4 Cold crossing tails modify the balance

Let $z(L) = \mathbb{E}G_{T_L}$ and $d = q(p_i - p_w)$. The complete physical rate is

$$j(L) = j_0(L) + \frac{d\theta z(L)}{U(L)}. \quad (18)$$

The terminal tail is consumed by the compactor at input price rather than written by the next ordinary request. Consequently, the correction depends on $p_i - p_w$, not $p_i - p_r$.

At smooth points define $\chi = z - Uz'/u$. The stationary balance becomes

$$\Phi(L) = cD(L) - A - d\theta\chi(L) = 0. \quad (19)$$

For exponential growth, $\chi = g[2 - (2 + L/g)e^{-L/g}]$ and $\chi' > 0$. Under $p_i \leq p_w$, $\Phi' = cU - d\theta\chi' > 0$, giving a unique marked crossing. At $q = 1$, the write-price derivative has the sign of

$$(S - B) - \theta\chi(L_*). \quad (20)$$

Thus a cold tail can overturn the core’s positive write-price response when volatile reconstruction is thin. This mechanism distinguishes the existence of a sign reversal from its economic importance. For empirical integer laws we evaluate the marked ledger directly and compare optimal cells.

3.5 Invoice geometry and the value of retuning

The threshold results live within a broader price geometry. For a fixed workload law and feasible policy class, let $\mathbf{m}_\pi$ be expected disjoint token use under policy π . This may include history-dependent restoration and finite task endings. The optimized invoice is

$$V(\mathbf{p}) = \inf_{\pi} \mathbf{p}^T \mathbf{m}_\pi. \quad (21)$$

It is concave, coordinatewise nondecreasing, and homogeneous of degree one. At a differentiable attained optimum, $\nabla V = \mathbf{m}_{\pi_*}$; at a tie, active use vectors support the concave envelope [12]. Invoice price elasticities are nonnegative expenditure shares that sum to one. Threshold price elasticities, in contrast, can be negative and sum to zero.

Raising one price and reoptimizing weakly reduces use of that priced resource. This revealed-preference statement orders resource use even when the threshold moves in either direction. Appendix D supplies the short proof.

For the smooth core, set $\mathbf{w} = \mathbf{a} - R\mathbf{v}$. At an interior optimum with $u_* > 0$,

$$\nabla_{\mathbf{p}}^2 j_0^* = -\frac{u_*}{cU_*^3} \mathbf{w} \mathbf{w}^T. \quad (22)$$

The curvature has rank at most one. Retaining the old threshold after a small price change $\delta \mathbf{p}$ incurs local excess cost

$$\text{regret} \simeq \frac{u_*}{2cU_*^3} (\mathbf{w}^T \delta \mathbf{p})^2. \quad (23)$$

A threshold can move at first order while the benefit of retuning is second order. For atomic workloads, exact price-linear switch margins take the place of smooth curvature. Both views favor measuring decision loss rather than interpreting every change in the argmin as an operationally significant event.

4 Evaluation

Our evaluation asks how the theoretical mechanisms behave under a measured growth law: whether the renewal prediction survives large increments and local dependence, when price effects reverse, and how much money a changed controller actually saves. We estimate growth from community traces and intervene on prices, reconstruction, cache reliability, and timing separately.

4.1 Estimating an engineering continuation workload

We use the public TraceLab v0.0.2 release [24], containing 665,453 sanitized model-call records. We select 305,445 Claude calls, group them into 5,320 session/file sequences within provider and project, and order by round index and ingestion ordinal. Retained growth is estimated as the difference between consecutive full input lengths:

$$\widehat{G}_t = \text{fullInput}_{t+1} - \text{fullInput}_t. \quad (24)$$

Full input is the relevant physical history quantity; fresh billable input also depends on cache replay. The estimator includes intervening output and tool payloads, which avoids counting output again as additional growth.

We form a monotone-continuation stratum by retaining nonnegative consecutive changes outside the three transitions following a drop of at least 64k tokens. Negative transitions are excluded rather than clipped. The resulting pool contains **297,091** increments. This extraction isolates the growing segments to which the regenerative analysis applies. The drop marker is a segmentation heuristic; reconstruction and cache parameters are supplied separately rather than inferred from an ambiguous drop label.

The full accepted pool has mean 1,836 tokens, median 909, and a maximum above 310k. Its largest 1% contributes 57.6% of the second raw moment and 95.7% of the third. These jumps motivate solving the empirical renewal convolution instead of selecting a smooth family from a mean and variance. A threshold around 100k is not automatically large relative to this increment law.

A matched dependence control. We sample intact 16-action moving blocks within uninterrupted runs. To compare dependence rather than different cohorts, the iid control uses the exact one-step marginal induced by those blocks. A uniform random initial phase in $\{0, \dots, 15\}$ makes the marginal equal at every action index, including the finite start and end. This matched marginal has $g = 1,532.8794$ tokens. Chronologically aligned billed output marks

have mean $o = 848.0400$ and stay paired with growth. The block control retains local clustering while the iid control removes it. Details of the weighting and exclusions are in Appendix E.

Reconstruction and cache interventions. The baseline uses $S = 65,588$ and $C = 4,382$, magnitude anchors from an independent coding-agent case study [11]. The stable decomposition $B = 20,000$ is a controlled within- S parameter, with $K = I = 128$ and $\theta = 1$. We vary these quantities along explicit material paths. Cache probabilities are controlled conditions $q = .8, .95, 1$, allowing the price effects to be separated from the reliability of a particular deployment.

4.2 Analytical selection and trajectory confirmation

The plug-in empirical law is evaluated through deterministic renewal convolution on a 5-token lattice. A 25-token check moves the baseline marked selection by 15 tokens and its rate by 4.11×10^{-7} currency/action. The analytical choice includes the cold terminal mark and a right-tail certificate over the unbounded threshold domain for the finite binned law.

Trajectory comparisons use raw integer increments. Settings share growth, output, and cache fields indexed by ordinary action. We simulate 512 trajectories with horizons up to 4,000 actions. The initial context is warm and the invoice stops with the final ordinary action. Short-horizon exploratory choices are checked using independently seeded trajectories. Price experiments simulate one four-category ledger per mechanism and reprice it across all vectors, yielding paired comparisons without resampling work. Reported intervals are conditional 95% Monte Carlo intervals over this empirical workload. Appendix M records the seeds, numerical checks, and artifact mapping.

4.3 A broad minimum, with variance shaped by dependence

For baseline prices (3, 3.75, .30, 15) per million tokens, the empirical core selects 113,190 tokens and the marked ledger selects **112,848**, at a long-run rate of 0.052701 per ordinary action. The fluid leading term selects 115,428. A second-moment correction gives 112,780, while a third-moment correction gives 113,511. More moments do not produce a monotonic improvement: the higher moments are especially exposed to rare large observations.

Figure 2 shows the central distinction between the location of a minimum and the significance of its neighborhood. In the 4,000-action workload, sampled thresholds around 106k–126k lie within 1% of the lowest mean invoice. Both exploratory dependence controls selected 113,848, only 1k above the analytical choice; independent confirmation leaves their small invoice differences unresolved. The analytical crossing supplies a useful center without demanding a token-precise operational setting.

Dependence is more visible in risk than in the mean minimum. Disjoint 16-action growth sums have 3.16 times the variance of matched iid sums. At the analytical threshold, 4,000-action block invoice-rate variance is 4.14 times the iid value. The near-coincident mean curves therefore coexist with substantially different budget

Table 1: Material paths from the baseline. Savings are confirmed block retuning gains relative to keeping the original 112,848-token controller.

Intervention	New H (k)	Savings
Existing 20k becomes stable	105.2	0.35%
Add 20k stable material	133.5	3.55%
Add 20k volatile material	140.0	5.85%
Summary grows to 20k, preserve material	152.7	9.82%
Scale the whole growth law by two	130.8	1.50%

variability. A controller tuned only to the mean threshold would miss this workload feature.

With 32 ordinary actions remaining, higher thresholds selected on exploratory trajectories reduce independently confirmed expense by 4.78% under iid work and 3.94% under blocks. Their paired improvements are 0.002269 ± 0.000268 and 0.001824 ± 0.000234 per action. Here the finite ending removes enough future reuse to make resetting harder to amortize. Remaining work changes the economics more than the 1k adjustment inside the long-run basin.

4.4 Predicted reversals survive empirical growth

At the baseline anchors, Proposition 3.2 gives a core crossover $q_w = 0.885744$. Figure 3 evaluates the full marked response to a 5% write-price increase. The empirical finite-change boundary lies near $q = .90$: the cold-tail ledger and discrete cells shift the core prediction, while retaining its central mechanism.

A 20% write-price increase moves the main $q = 1$ threshold from 112.8k to 115.9k, but moves the $q = .8$ threshold from 91.0k to 90.4k. Each comparison uses its own unchanged-price, same-mechanism controller, so the observed price effect is separated from the much larger effect of changing q .

We then isolate the cold-tail reversal with $B = 64,000$, $C = 100$, and the same full $S = 65,588$. At $q = 1$, raising write price by 20% moves the marked threshold *earlier*, from 79,368 to 78,893, while the core predicts a later crossing. Moving the same growth to pre-call input ($\theta = 0$) restores the later response, from 80,248 to 80,603. This paired timing intervention identifies the mechanism in Equation (20). Its old-policy excess is only about 0.0052%, illustrating why the sign of a response and the financial value of acting on it deserve separate attention.

4.5 Working-state design outweighs fine retuning

Reclassifying 20k existing tokens as stable reduces repeated reset writes and moves the optimum to 105.2k. Adding 20k stable tokens instead raises the reconstructed floor and moves it to 133.5k. Adding volatile material raises both the floor and rewrite burden, selecting 140.0k. Increasing the summary to 20k while preserving other material selects 152.7k. Table 1 shows that failing to adapt to a changed working state can cost materially more than ignoring a modest single-price adjustment. The experiments isolate the invoice consequences of these representations; their informational benefits are a separate design input.

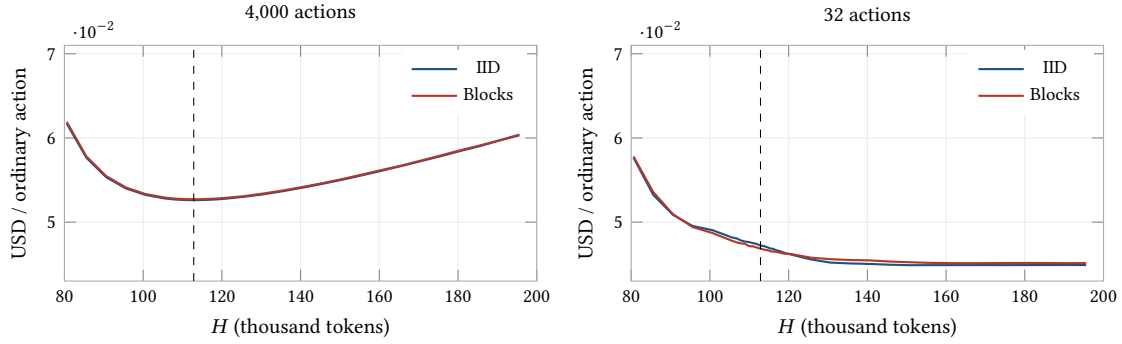


Figure 2: Community-derived invoice curves under matched marginals. Left: long-horizon iid and block means track the analytical minimum; the bottom is broad. Right: with only 32 actions remaining, the warm initial state makes later compaction more attractive. Lines join evaluated points; the vertical line marks the long-run analytical choice.

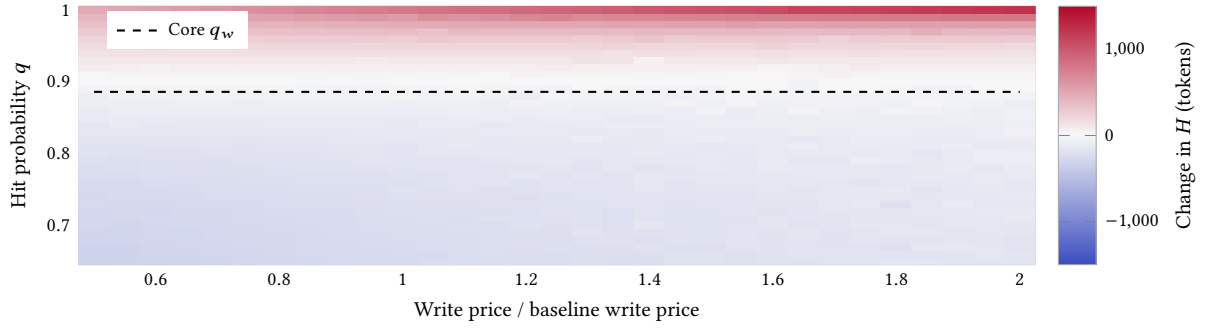


Figure 3: Full marked threshold response to a 5% write-price increase on community increments. Blue favors earlier compaction and red later compaction. The dashed line is the core theory’s q_w , which is independent of write price. The marked empirical boundary reflects cold-tail and cell effects.

4.6 Restoration semantics change the decision itself

The material-path contrasts hold an aggregate reconstruction protocol fixed. We next test the premise behind that reduction: does the required state arrive immediately after a reset, or only when a subsequent operation needs it? An earlier, complementary versioned-file experiment separates those mechanisms. It uses measured whole-file sizes and community output blocks, while explicitly controlling file demand, revisions and request gaps. Unlike the full-input renewal experiment, it constructs file input through the simulator; adding observed full-input increments would count restoration twice. Consequently its absolute costs and thresholds are interpreted within its own workload, not pooled with the renewal estimates.

Four pinned files from Jinja, Rich, CPython and Django contain 61,318 tokens under a reproducible tokenizer proxy. With a 20k stable base, a 4,382-token summary and observation wrappers, eager reconstruction creates an 85,956-token state. A 400-action phase-local workload asks for at most one file per action, with a .85 requirement probability, .9 preference for the current hot file, 25-action phases and .002 revision probability per file/action. Request gaps have a declared 90-second mean and coefficient of variation

1.5; cache lifetime is 300 seconds. The four billing categories price ordinary requests, compactors and optional model-mediated recovery uniformly.

At independently confirmed, discovery-selected coarse thresholds, eager reconstruction with a surviving stable base costs \$30.6022 per task, whereas first-use reconstruction costs \$22.9665. The paired reduction is \$7.6357, with a conditional 95% Monte Carlo interval of [7.5923, 7.6792], or about 24.95% of the eager bill. First use restores *more* payload—801k rather than 690k tokens—but pays less for repeated full-context writes and reads. Retaining file observations and their read guards restores only 587k tokens yet costs \$24.4286. Reload volume alone therefore orders these policies incorrectly.

New 4,096-trajectory discovery and confirmation samples refine the phase-local line to 78k, with a sampled 1% region of 68–86k, and the all-files-mandatory line to 126k, with a region of 116–138k (Figure 4). Confirmed task bills are respectively \$22.9543 $\pm$.0508 and \$30.7784 $\pm$.0665. Under mandatory demand, eager, first-use and valid-guard retention have identical complete invoices in this engine: all must supply the same next input. Concentrated compulsory reconstruction is thus a meaningful limiting protocol, rather than a justification for ignoring first-use semantics elsewhere.

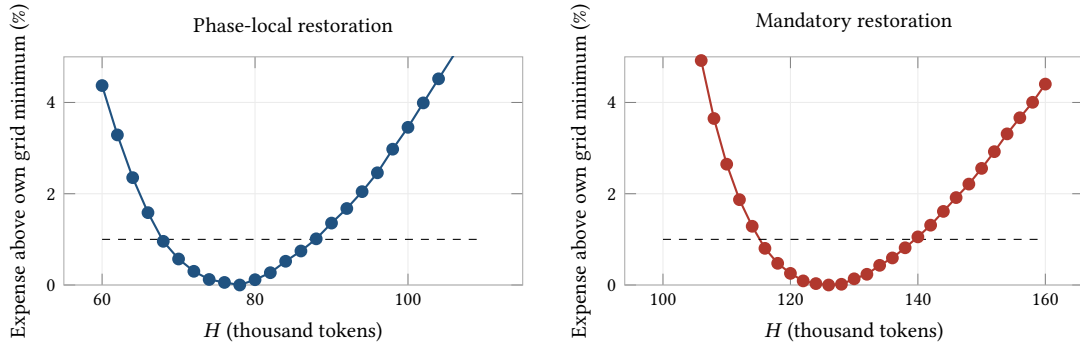


Figure 4: Separately executed versioned-file fine-grid experiments. Each curve is normalized by its own sampled discovery minimum; the dashed line marks 1% excess expense. Phase-local first-use restoration selects 78k, whereas requiring all files at every action selects 126k. Independent confirmation freezes those choices. These are different required-state laws, not competing estimates of the same universal threshold.

The distinction also changes operational comparisons. Diffuse demand reduces first use’s benefit; delaying both compaction and recovery by 360 seconds erodes it through expiry; forgetting retained read guards adds duplicate input. A jointly selected retention-budget experiment finds that partial retention can cost more than either dropping all files or retaining the entire catalog when extra recovery inference expires the cache. In an exact two-file control, batching an additional recovery model request reverses eager versus first use. Appendix F presents the policy and intervention matrices; Appendix G isolates the finite mechanisms. Together they locate the scalar renewal model’s useful domain: a prompt floor must describe genuinely recurring working state, not merely text that used to be present.

4.7 Price families and cache-mode trade-offs

We collected 38 first-party quoted price vectors on October 6, 2026: 17 GPT/OpenAI entries, 12 Qwen entries, four DeepSeek time-band entries, and five Claude entries [3, 7, 14, 19–21]. Together with the retained mechanism controls, the three hit conditions produce 143 analytical scenarios and paired trajectory confirmations. Tariffs are applied to the same token ledger, exposing price geometry independently of model generation and tokenization.

Several nominally different prices share one control class. GPT-6 Astra, GPT-6 Sol, GPT-6 Luna, and GPT-5.6 Sol short-context vectors normalize to (1, 1.25, .10, 5) and select the same 112.8k line at $q = 1$. GPT-6.1 Sol’s relative read price is .05 and selects 132.8k; GPT-4.1’s .25 selects 92.4k. The comparative structure is relative, not a monotone ordering by the absolute input price.

DeepSeek Flash’s off-peak vector (.15, .15, .003, .6) makes reads exceptionally cheap, selecting 162.0k at $q = 1$. At $q = .95$ it selects 116.9k, and at $q = .8$ 94.4k. Cheap reads amplify the value of reliable reuse rather than removing the need to measure reliability. Peak prices double every component and leave these controllers unchanged. Twenty-seven proportional-vector controls across the three q values reproduce identical selections and proportionally scaled analytical and simulated invoices.

Qwen provides a complementary contrast. For 3.8 Max in Singapore, explicit caching quotes write/read prices of 2.5/.17, versus

2/.25 for implicit caching, with the same ordinary input and output prices. At matched $q = .8$, the optimized explicit projection costs 6.87% more; at $q = .95$ it costs 6.51% less; and at $q = 1$ it costs 15.97% less. Paired block total-expense differences are respectively $+.004271 \pm .000028$, $-.002688 \pm .000020$, and $-.005446 \pm .0000085$ per action. Repeated discounted reads eventually repay the creation premium. A real mode comparison can additionally change reliability, which the matched- q experiment keeps separate.

An analogous cache-lifetime tariff decision benefits from a break-even calculation. With Sonnet 5.5’s short tariff at $q = .8$, its higher-write one-hour tariff needs $q \approx .8950$ after reoptimization to break even. The declared improvement to $q = .95$ saves about 23.3% in block confirmation; changing only the tariff while keeping $q = .8$ costs about 38.4% more. The useful deployment question is whether that reliability improvement actually occurs, rather than whether a longer advertised lifetime sounds economical.

5 Discussion

5.1 Compaction is state reconstruction, not text deletion

The findings suggest treating a compacted context as a *restartable working state*. A summary records decisions; source files and tools provide exact facts; a stable prefix preserves computational reuse. These roles are complementary. Summarizing a file’s contents can remove prompt text without removing the next operation’s need to inspect the file. Conversely, a durable artifact handle can defer restoration until first use. Working-state design determines what S , B , and C mean, and therefore determines which side of the price balance a change affects.

The contrast between stable reclassification and stable addition is especially instructive. One improves the reuse of already required information; the other enlarges the information carried after every reset. A design discussion framed only as “keep more context” merges these different interventions. The appropriate unit of optimization is a representation plus its reconstruction contract, followed by a control policy for that representation.

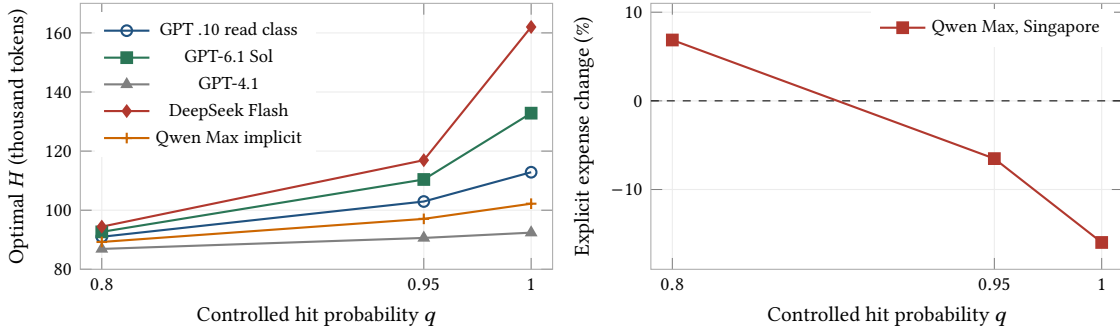


Figure 5: Quoted-vector projections at controlled cache reliabilities. Left: very cheap reads support substantially larger contexts near perfect reuse, but the difference contracts with misses. Right: Qwen Max’s explicit write/read trade-off changes sign even when both modes share the same q . Markers are evaluated conditions; connecting lines guide the eye.

5.2 Optimize the loss basin before the trigger point

The rank-one curvature explains why four separate price sensitivities need not imply four independently valuable tuning axes. Locally, only the price direction that changes the balance between reconstruction and carrying moves a smooth scalar controller. Common price scaling lies in a null direction. A modest displacement along a shallow basin can be less valuable than a layout change that preserves a reusable prefix or a timing change that improves hit probability.

An economical operational order follows: establish the working state and cache boundaries, estimate reuse reliability, locate the renewal crossing, and then choose a robust threshold region. Measure retuning regret, not just the distance between two point estimates. Atomic plateaus make this advice stronger: many threshold values generate the same schedule and invoice. For short tasks, the remaining-work hazard is a higher-value control signal because it determines whether a reset has time to repay itself.

5.3 Mean economics and budget risk require different summaries

Local clustering raises budget variability even when the mean cost basin changes little. This creates a second decision layer: mean-optimal compaction and budget-constrained execution need not select the same operating region. A risk-sensitive controller could track remaining budget together with working-state size and phase. The empirical block results make that direction concrete, while the price-envelope identity continues to provide resource attribution for any fixed policy. Estimating phase-conditioned occupations is more informative here than fitting several independent noise distributions.

5.4 From cheaper continuation to completed tasks

The fundamental objective question is not whether a shorter context can lower an invoice, but whether the saved resources still buy a completed task. The experiments fix ordinary work and its exogenous observations. Finishing 400 simulated actions establishes

ledger completion, not that a coding agent has fixed an issue. An agent can cheaply repeat actions, lose an obligation in a summary, or stop without a valid patch. Conversely, reconstruction can be expensive but necessary for a correct solution. This distinction changes the optimization target, not just the width of an uncertainty interval.

For a fixed task population and evaluation contract, let $Y_\pi \in \{0, 1\}$ indicate externally verified success, ℓ_π the random four-category usage over the entire attempt, and $Z_\pi = \mathbf{1}^\top \ell_\pi$ the total billed tokens. Failed attempts, recovery requests, summaries and retries remain in the denominator. The operational efficiencies are

$$\eta_T(\pi) = \frac{\mathbb{E}Y_\pi}{\mathbb{E}Z_\pi}, \quad \eta_S(\pi) = \frac{\mathbb{E}Y_\pi}{\mathbf{p}^\top \mathbb{E}\ell_\pi}. \quad (25)$$

They are completed tasks per billed token and per dollar, respectively. With repeated identically distributed attempts and suitable integrability, total successes divided by total resources converges to these ratios of expectations. Averaging individual Y/Z ratios instead gives a different weighting, while averaging tokens only over successful runs omits the price of failure. The task distribution must remain fixed: replacing difficult issues with easy ones can improve the ratio without improving the agent.

Token efficiency and dollar efficiency answer related but distinct questions. One cache-read token and one generated token contribute equally to Z but have different prices. Both should be reported alongside success rate and the category ledger. Execution-based benchmarks such as SWE-bench anchor success in an accepted patch and tests [27]; ordinary-call counts cannot stand in for that contract. Long-context retrieval evidence also shows that retaining more text does not monotonically improve use of relevant information [26]. Shortening may reduce distraction, or it may remove the precise fact needed later. The relevant mechanism is information availability and use, not length alone.

This yields a sharp decision boundary. For policies A, B with positive success probabilities s_A, s_B and expected bills C_A, C_B , a cheaper A is less dollar-efficient whenever

$$\frac{s_A}{s_B} < \frac{C_A}{C_B}. \quad (26)$$

A 20% bill saving can tolerate at most a 20% *relative* success loss before losing throughput; the corresponding token boundary replaces

C by $\mathbb{E}Z$. This is an algebraic trade-off, not an estimated quality curve for our community pool. Because that pool supplies usage without policy-intervention success labels, it identifies the resource side of this boundary, not the full task-efficiency optimum.

There are two practical next objectives. A reliability-sensitive workflow minimizes $\mathbf{p}^\top \mathbb{E}\mathbf{l}_\pi$ subject to $\mathbb{E}Y_\pi \geq \tau$, for a declared success floor τ . A throughput-sensitive workflow maximizes η_T or η_S and reports the resulting success level. Ratio optimization has a useful fractional-programming form [25]: at the optimal dollar efficiency λ , the best policy satisfies

$$\max_{\pi} \{\mathbb{E}Y_\pi - \lambda \mathbf{p}^\top \mathbb{E}\mathbf{l}_\pi\} = 0. \quad (27)$$

The threshold responds to both resource and success gradients; a cost-only stationary point is generally not a task-efficiency stationary point. Appendix L derives these statements and a task-level measurement design. The present cost geometry remains a component of that larger problem: it attributes the invoice and identifies which mechanisms deserve intervention, while the successful-task frontier decides whether an economical state is actually useful.

5.5 Length is a useful reduction, not a sufficient state

The restoration experiments suggest an adaptive-control boundary with an exact counterpart. Two reachable states can have the same 10k context, cache state, one remaining action and future requirement probability, but differ in whether a current-version file is visible. In a cold-prefix control, the valid-file state costs \$.0379215 to continue and \$.0401805 to compact; the absent-file state costs \$.0454215 to continue and the same \$.0401805 to compact. Their strict optimal actions are opposite. The difference is the option destroyed by compacting a valid fact, not an unmodeled random reset fee.

For an equal mixture, the best state-aware bill is \$.039051; the best length-only policy costs \$.0401805. Even randomizing the length-only action leaves unavoidable regret \$.0011295, or 2.81% of that restricted bill. Appendix G gives reachable histories, all-four-active ledger checks, and matching exact Bellman and occupation-flow certificates. This establishes a structural limitation, rather than extrapolating a simulation curve to all tasks. A scalar line is useful when reconstruction and reuse conditions are sufficiently stable; validity, phase and remaining work become valuable control signals when they alter the future invoice.

5.6 Scope and limitations

The evaluated objective is invoice cost for fixed ordinary work and a supplied working-state representation. Compression can change task success, retries, and future observations; joint utility-aware control requires those responses in the transition law. The concentrated reset model describes phases with a recurring aggregate state. Versioned artifacts, delayed first use, additional recovery inference, and changing phases introduce state variables beyond context length. In that setting, equal-length histories can require different actions, and the scalar threshold becomes a policy restriction.

The growth estimator is a policy-conditioned public-trace stratum. Aggregate usage lacks explicit artifact identities and reliable compact labels, so reconstruction composition, retained-tail timing,

and cache reliability enter as interventions. The independently observed S, C anchors come from one practitioner’s case study; their purpose is to instantiate engineering-scale state, not to estimate a population optimum. Block resampling retains local dependence rather than complete task evolution. Conditional Monte Carlo intervals describe simulated variability given that pool, while corpus selection and model behavior remain additional sources of uncertainty.

Price vectors are constant-tariff projections. Actual GPT and Qwen long-input prices are selected per request, including a potentially oversized compactor request; DeepSeek time bands depend on execution time. A band-conditioned occupation ledger handles these distinctions, whereas selecting one vector from H alone does not. Specific Qwen Flash creation quotes also differ from generic multiplier prose; the catalog records the specific table and its discrepancy. Real service comparisons additionally require model-specific tokenization, cache eligibility, latency, and output behavior. These extensions preserve the fixed-policy price geometry but can change the threshold balance.

6 Conclusion

The useful compact line is an amortization decision: how long to keep paying for the current history before purchasing a reconstructed state. Integrated renewal occupation turns that decision into a distribution-aware crossing, and the four-price ledger reveals when familiar intuitions reverse. The community experiments show where this structure has practical force: cache reliability can erase the apparent advantage of extremely cheap reads, working-state changes can dominate small price retuning, and finite endings can favor postponement even when steady operation favors cycling.

The next step is to estimate the joint law of eligible-prefix length, cache age, restoration demand, and remaining work, then use it to choose when and what to reconstruct. Request-band prices and task utility can be incorporated at that state level. The present analysis provides the organizing principle: preserve the distinction between information available to the agent, computation reusable by the service, and work left to amortize the next reset.

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References

- [1] Anthropic. 2025. Effective Context Engineering for AI Agents. <https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents>.
- [2] Anthropic. 2026. Prompt Caching. Accessed October 6, 2026. <https://platform.claude.com/docs/en/build-with-claude/prompt-caching>.
- [3] Anthropic. 2026. Pricing. Accessed October 6, 2026. <https://platform.claude.com/docs/en/about-claude/pricing>.
- [4] Peter J. Denning. 1968. The Working Set Model for Program Behavior. *Communications of the ACM* 11, 5 (1968), 323–333. <https://doi.org/10.1145/363095.363141>.
- [5] John T. Daly. 2006. A Higher Order Estimate of the Optimum Checkpoint Interval for Restart Dumps. *Future Generation Computer Systems* 22, 3 (2006), 303–312. <https://doi.org/10.1016/j.future.2004.11.016>.
- [6] DeepSeek. 2026. Context Caching. Accessed October 6, 2026. https://api-docs.deepseek.com/guides/kv_cache/.
- [7] DeepSeek. 2026. Models & Pricing. Accessed October 6, 2026. https://api-docs.deepseek.com/quick_start/pricing/.
- [8] Huiqiang Jiang, Qianhui Wu, Chin-Yew Lin, Yuqing Yang, and Lili Qiu. 2023. LLMingua: Compressing Prompts for Accelerated Inference of Large Language

- Models. In *EMNLP*. 13358–13376. <https://doi.org/10.18653/v1/2023.emnlp-main.825>.
- [9] Tobias Lindenbauer, Igor Slinko, Ludwig Felder, Egor Bogomolov, and Yaroslav Zharov. 2025. The Complexity Trap: Simple Observation Masking Is as Efficient as LLM Summarization for Agent Context Management. *arXiv:2508.21433*. <https://arxiv.org/abs/2508.21433>.
- [10] Elias Lumer, Faheem Nizar, Akshaya Jangiti, Kevin Frank, Anmol Gulati, Mandar Phadate, and Vamse Kumar Subbiah. 2026. Don’t Break the Cache: An Evaluation of Prompt Caching for Long-Horizon Agentic Tasks. *arXiv:2601.06007*. <https://arxiv.org/abs/2601.06007>.
- [11] LangWatch. 2026. Finding the Optimal Context Window for Coding Agents. <https://langwatch.ai/research/finding-the-optimal-context-window>.
- [12] Paul Milgrom and Ilya Segal. 2002. Envelope Theorems for Arbitrary Choice Sets. *Econometrica* 70, 2 (2002), 583–601. <https://doi.org/10.1111/1468-0262.00296>.
- [13] OpenAI. 2026. Prompt Caching. Accessed October 6, 2026. <https://developers.openai.com/api/docs/guides/prompt-caching>.
- [14] OpenAI. 2026. API Pricing. Accessed October 6, 2026. <https://developers.openai.com/api/docs/pricing>.
- [15] Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. 2023. MemGPT: Towards LLMs as Operating Systems. *arXiv:2310.08560*. <https://arxiv.org/abs/2310.08560>.
- [16] Zhuoshi Pan, Qianhui Wu, Huiqiang Jiang, Menglin Xia, Xufang Luo, Jue Zhang, Qingwei Lin, Victor Rühle, Yuqing Yang, Chin-Yew Lin, H. Vicky Zhao, Lili Qiu, and Dongmei Zhang. 2024. LLMingua-2: Data Distillation for Efficient and Faithful Task-Agnostic Prompt Compression. In *Findings of ACL*. 963–981. <https://doi.org/10.18653/v1/2024.findings-acl.57>.
- [17] Ritul Satish, Prasoon Sinha, Akiho Kawada, and Neeraja J. Yadwadkar. 2026. Beyond Token Savings: A Systematic Study of Context Compression in LLM Agents. *arXiv:2609.32961*. <https://arxiv.org/abs/2609.32961>.
- [18] Alibaba Cloud. 2026. Context Cache. Accessed October 6, 2026. <https://www.alibabacloud.com/help/en/model-studio/context-cache>.
- [19] Alibaba Cloud. 2026. Qwen3.8-Max Model Information. Accessed October 6, 2026. <https://www.alibabacloud.com/help/en/model-studio/qwen3-8-max>.
- [20] Alibaba Cloud. 2026. Qwen3.8-Flash Model Information. Accessed October 6, 2026. <https://www.alibabacloud.com/help/en/model-studio/qwen3-8-flash>.
- [21] Alibaba Cloud. 2026. Qwen3.7-Plus Model Information. Accessed October 6, 2026. <https://www.alibabacloud.com/help/en/model-studio/qwen3-7-plus>.
- [22] Buqiang Xu, Zirui Xue, Dianmou Chen, Chenyang Fu, Chiyu Wu, Caiying Huang, Chen Jiang, Jizhan Fang, Xinle Deng, Yijun Chen, Yunzhi Yao, Xuehai Wang, Jin Shang, Gong Yu, and Ningyu Zhang. 2026. TokenPilot: Cache-Efficient Context Management for LLM Agents. *arXiv:2606.17016*. <https://arxiv.org/abs/2606.17016>.
- [23] John Yang, Carlos E. Jimenez, Alexander Wettig, Kilian Lieret, Shunyu Yao, Karthik Narasimhan, and Ofir Press. 2024. SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering. In *Advances in Neural Information Processing Systems* 37. https://papers.nips.cc/paper_files/paper/2024/hash/5a7c947568c1b1328ccc5230172e1e7c-Abstract-Conference.html.
- [24] Kan Zhu, Mathew Jacob, Chenxi Ma, Yi Pan, Stephanie Wang, Arvind Krishnamurthy, and Baris Kasikci. 2026. TraceLab: Characterizing Coding Agent Workloads for LLM Serving. *arXiv:2606.30560*. <https://arxiv.org/abs/2606.30560>.
- [25] Werner Dinkelbach. 1967. On Nonlinear Fractional Programming. *Management Science* 13, 7, 492–498. <https://doi.org/10.1287/mnsc.13.7.492>.
- [26] Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. Lost in the Middle: How Language Models Use Long Contexts. *Transactions of the Association for Computational Linguistics* 12, 157–173. https://doi.org/10.1162/tacl_a_00638.
- [27] Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. 2024. SWE-bench: Can Language Models Resolve Real-World GitHub Issues? In *International Conference on Learning Representations*. https://proceedings.iclr.cc/paper_files/paper/2024/file/edac78c3e300629acf6cbe9ca88fb84-Paper-Conference.pdf.

A Renewal Accounting and the Crossing Theorem

This appendix derives the rate and the distribution-general crossing. All thresholds satisfy $L > 0$, with iid $G \geq 0$ and $0 < \mathbb{E}G = g < \infty$. There exist $\epsilon > 0$ and $\rho > 0$ such that $\mathbb{P}(G \geq \epsilon) = \rho$. The time to accumulate $\lfloor L/\epsilon \rfloor$ such successes bounds T_L from above by a negative-binomial waiting time. Thus $\mathbb{E}T_L < \infty$ and the renewal measure is locally finite. When $p_0 = \mathbb{P}(G = 0) > 0$, its mass at zero is $1/(1 - p_0)$, accounting for repeated zero-growth actions.

A.1 Occupation and overshoot

Nonnegative increments imply $\{T_L > n\} = \{Y_n < L\}$. Tonelli’s theorem gives

$$\mathbb{E}T_L = \sum_{n \geq 0} \mathbb{P}(Y_n < L) = U(L), \quad (28)$$

$$\mathbb{E} \sum_{n=0}^{T_L-1} Y_n = M(L). \quad (29)$$

Because $\{T_L \geq n\}$ is determined before G_n , the stopped-sum identity is

$$\mathbb{E}Y_{T_L} = \sum_{n \geq 1} \mathbb{E}[G_n \mathbf{1}\{T_L \geq n\}] = gU(L). \quad (30)$$

Hence affine charges for newly accumulated crossed history contribute a threshold-independent rate κg . Reprocessing S contributes $\kappa S/U$ and remains threshold-sensitive. This separates a recurrent floor from new history processed once at the end of each cycle.

Let $\mathcal{U}$ be the renewal measure. Then

$$D(L) = \int_{[0,L)} (L-x) \mathcal{U}(dx) = \int_0^L U(t) dt. \quad (31)$$

D is continuous, convex, strictly increasing, and unbounded because $U(t) \geq \mathcal{U}(\{0\}) \geq 1$. Its derivative is U at continuity points.

A.2 Four-category reconciliation

At one boundary the same hit mark determines old-prefix reuse for the compactor and, after a reset, the eligible prefix B of the following action. We average those marks before forming the expected cycle account, leaving the renewal calculation in the iid growth/output variables. Write $z = \mathbb{E}G_{T_L}$, $U = U(L)$, and $M = M(L)$. Expected token rates are

$$m_i = I + (1-q)g + \frac{K + (1-q)S + q\theta z}{U}, \quad (32)$$

$$m_w = [1 - (1-q)\theta]g + (1-q)S + \frac{q(S-B) + (1-q)M - q\theta z}{U}, \quad (33)$$

$$m_r = q \left[(1-\theta)g + S + \frac{B+M}{U} \right], \quad (34)$$

$$m_o = o + \frac{C}{U}. \quad (35)$$

These are expected physical categories and do not depend on prices. The negative separated terminal term in m_w subtracts a write that is avoided; the complete write count is nonnegative. Expanding $\mathbf{p}^T \mathbf{m}$ yields Equations (7) and (18) with the coefficients in Equations (5)–(6). Independent hits are averaged only after eligible lengths and cold tails are assigned their categories.

A.3 Proof of Theorem 3.1

At a density point, $U' = u$ and $M' = Lu$, so

$$\frac{d}{dL} \frac{A + cM}{U} = \frac{u}{U^2} (cLU - A - cM) = \frac{u}{U^2} (cD - A). \quad (36)$$

At an atom x of renewal mass w , take strict-before-boundary $U_- = U(x)$ and $M_- = M(x)$. Passing the atom adds w and xw :

$$j_0(x+) - j_0(x) = \frac{w}{U_-(U_- + w)} [cD(x) - A]. \quad (37)$$

For continuous components, the bounded-variation quotient rule gives the same signed change with respect to the non-atomic renewal measure. Since D is strictly increasing and unbounded, $cD - A$ crosses zero exactly once. All invoice changes before that crossing are nonpositive and all changes after it are nonnegative. Regions with no new renewal mass are constant-cost plateaus; the canonical $D^{-1}(A/c)$ selects a representative in the minimum region. This completes the measure-form proof.

B Comparative Statics of the Growth and Reset Laws

B.1 Growth scale

For scaled growth aG , $U_a(L) = U(L/a)$ and $D_a(L) = aD(L/a)$. Let $x = L_*(a)/a$, so $aD(x) = R$. Differentiating at a continuity point gives

$$\frac{dL_*}{da} = x - \frac{D(x)}{U(x)} = \frac{M(x)}{U(x)} \geq 0. \quad (38)$$

Consequently $\epsilon_{L,a} = M/(LU)$, complementary to $\epsilon_{L,R} = D/(LU)$. Both lie between zero and one.

B.2 Convex-order dispersion

If $G_1 \leq_{cx} G_2$ with equal means, independent sums preserve convex order. Since $y \mapsto (L-y)_+$ is convex, $\mathbb{E}[(L-Y_n^{(1)})_+] \leq \mathbb{E}[(L-Y_n^{(2)})_+]$ for every n . Summing nonnegative terms yields $D_1(L) \leq D_2(L)$. Inversion at the same R gives $L_*^{(2)} \leq L_*^{(1)}$. Moment or coefficient-of-variation differences alone need not impose convex order. Closure under independent sums follows by conditioning: both $x \mapsto \mathbb{E}\varphi(x+G)$ and $g_0 \mapsto \mathbb{E}\varphi(Y+g_0)$ preserve convexity for convex φ . Replacing independent components one at a time proves the inequality.

An exact counterexample distinguishes this ordering from a CV slogan. Law 1 has values 0, 2 with equal probabilities, mean one and CV one. Law 2 has .5 with probability .9 and 5.5 with probability .1, also mean one but CV 1.5. At $R = .1$, the crossing gaps are .05 and .1: the higher-CV law chooses a larger gap. At $R = 5$, law 1 chooses 2.25, while law 2 chooses $2 + (5 - 4.5245)/4.0951 \approx 2.1161$, now smaller. Below its 5.5 jump, law 2 has renewal masses $.9^k$ at $k/2$; these give the displayed exact crossing. The D curves cross, so the laws are not convex-ordered. This is a distributional counterexample, not a claimed production traffic law.

B.3 Concentrated random reconstruction

For iid bounded reset sizes S independent of growth, with H above their support and reconstruction coefficient $A(S)$, differentiation of expected cycle charge over expected cycle length yields

$$\mathbb{E}[D(H-S)] = \mathbb{E}[A(S)]/c. \quad (39)$$

Indeed, the variable cycle charge is $N(H) = \mathbb{E}[A(S) + c\{SU(H-S) + M(H-S)\}]$ and mean duration $\bar{U}(H) = \mathbb{E}U(H-S)$. At smooth points, $N' = cH\mathbb{E}u(H-S)$ and $\bar{U}' = \mathbb{E}u(H-S)$. The quotient derivative has sign $c\mathbb{E}D(H-S) - \mathbb{E}A(S)$. This derives the mixture crossing before specializing a law; substituting only mean reset discards curvature in D . For exponential growth, $\mu_S = \mathbb{E}S$ and $\sigma_S^2 = \text{Var } S$ give

$$H_* = \mu_S - g + \sqrt{g^2 + 2g\mathbb{E}[A(S)]/c - \sigma_S^2}. \quad (40)$$

when this interior solution lies above the reset support. This identifies a controlled extension for concentrated reconstruction variability instead of introducing arbitrary broad reset distributions.

C Price Directions and Marked Corrections

C.1 Core inversion and phase boundaries

Since $R = A/c$, $R_{pj} = (a_j - Rv_j)/c$. Applying $D' = U$ proves Equation (12). With $T_0 = p_iK + p_oC$, the write sign numerator expands to

$$N_w(q) = S(p_r - p_i)q^2 + (T_0 + 2p_iS - p_rB)q - (T_0 + p_iS). \quad (41)$$

For $q \leq B/S$ it is nonpositive. On $[B/S, 1]$, $N'_w = p_r(2qS - B) + T_0 + 2p_iS(1-q) > 0$, and $N_w(1) = p_r(S-B) > 0$. This proves the unique boundary in Proposition 3.2.

The read-price sign numerator is

$$N_r = q\{p_w(B - qS) - p_i[K + (1-q)S] - p_oC\}. \quad (42)$$

For $p_w > p_i$, a positive response occurs precisely below

$$q_r = \frac{p_wB - p_i(K + S) - p_oC}{(p_w - p_i)S} \quad (43)$$

when its numerator is positive. Otherwise the response is nonpositive. The cache-reliability ratio derivative has numerator

$$N_q = (p_w - p_r)[p_w(S - B) + p_iK + p_oC] + p_r(p_w - p_i)S. \quad (44)$$

Under $p_w \geq p_i \geq p_r$, it is nonnegative. General mixed sensitivities for parameters s, t that leave the renewal law fixed are

$$L_{st} = \frac{R_{st}}{U_*} - \frac{u_* R_s R_t}{U_*^3}. \quad (45)$$

The second term accounts for the endogenous root movement.

At a positive renewal atom, inverse directional slopes use the appropriate left or right U . They are not one common classical gradient. For example, deterministic $G = 1$, $q = 0$, $S = 10$, $B = C = K = 0$, and $p_i = p_w = 1$ give $L_* = 4$, with $D'_- = 4$ and $D'_+ = 5$. Increasing write price moves the inverse argument left and gives directional derivative -2.5 , rather than the right-slope value -2 .

C.2 Terminal mark and the full stationary balance

Differentiating z/U gives $(uz - Uz')/U^2$. This yields $j' = u\Phi/U^2$ for Equation (19). For exponential growth,

$$z(L) = g(2 - e^{-L/g}), \quad (46)$$

$$\chi(L) = g[2 - (2 + L/g)e^{-L/g}], \quad (47)$$

$$\chi'(L) = (1 + L/g)e^{-L/g}. \quad (48)$$

Let $W_* = cU_* - d\theta\chi'_*$. Under the stated price order $W_* > 0$, and the full price derivative is

$$L_{pj} = \frac{A_{pj} + \theta d_{pj}\chi'_* - D_*c_{pj}}{W_*}. \quad (49)$$

At $q = 1$, $A_{p_w} = S - B$, $d_{p_w} = -1$, and $c_{p_w} = 0$, proving Equation (20). For other laws χ can be irregular; the empirical experiments use exact marked cell costs.

Expected eligible ordinary and compactor prefixes per cycle are

$$P_n = SU + M - \theta(gU - z) - (S - B), \quad (50)$$

$$P_c = S + gU - \theta z. \quad (51)$$

Thus the full fixed-threshold invoice derivative is

$$j_q = -\frac{(p_w - p_r)P_n + (p_i - p_r)P_c}{U} \leq 0 \quad (52)$$

when both input/write prices exceed the read price. Improved reliability reduces the bill even if marked threshold directions reverse.

D Envelope Geometry and Retuning Loss

An infimum of price-linear invoices is concave, common scaling factors out, and nonnegative use gives coordinatewise monotonicity. For a finite candidate set, a price-direction derivative at a tie is $\min_{\pi \text{ active}} \delta \mathbf{p}^T \mathbf{m}_{\pi}$. At a differentiable attained optimum it equals $\delta \mathbf{p}^T \mathbf{m}_{\pi^*}$.

For the own-resource ordering, let two policies have old-price bills b_0, b_1 and usage u_0, u_1 of the component raised by $\delta > 0$. Optimality gives $b_0 \leq b_1$ and $b_1 + \delta u_1 \leq b_0 + \delta u_0$. Combining them yields $\delta u_1 \leq \delta u_0$, hence $u_1 \leq u_0$. The theorem extends to arbitrarily correlated finite task weights. Its arithmetic implication is kernel-checked in the accompanying Lean source.

For the smooth core, $j_{LL} = cu/U$ at the root, and differentiation with respect to p_j gives $j_{Lp_j} = -uw_j/U^2$. Implicit optimization and the Schur complement therefore give

$$\nabla^2 j_0^* = -\frac{j_{pL} j_{Lp}}{j_{LL}} = -\frac{u}{cU^3} \mathbf{w} \mathbf{w}^T. \quad (53)$$

The old-policy excess follows by the quadratic expansion in threshold: $\frac{1}{2} j_{LL} (\delta L)^2$ with $\delta L = \mathbf{w}^T \delta \mathbf{p} / (cU)$, proving Equation (23). For discrete candidate policies, equality boundaries instead satisfy

$$\mathbf{p}^T (\mathbf{m}_h - \mathbf{m}_k) = 0, \quad (54)$$

and exact switch loss is price-linear. These finite price regions supply an alternative robustness description at atoms.

E Data Estimation and Experimental Detail

E.1 Source and extraction

The pinned [TraceLab v0.2 release](#) has compressed JSONL SHA-256

11ce51ec0a25e3d1d95b025bca2f7d16
47e47571eb7cc968acd5fc64d4b4fb65

Table 2 records every adjacent transition’s disposition.

Table 2: Chronological growth-cohort extraction.

Transition class	Count
Accepted monotone continuation	297,091
Negative input change	1,041
Recovery-window nonnegative change	1,993
Nonconsecutive round index	0
Total adjacent transitions	300,125

The detector finds 671 large-drop candidates. Grouping is by provider, project, session ID, and session file; each excluded transition breaks a run. The accepted pool has 5,836 uninterrupted runs

and 232,186 eligible length-16 moving-block starts. Candidate post-drop prompt sizes do not identify generated summaries or fully reconstructed working states, which is why those quantities are not fitted from this detector.

E.2 Matched marginal and output marks

Let $\mathcal{B}$ be the eligible moving blocks of length $\ell = 16$. The marginal weight of observation j is

$$w_j = \frac{1}{\ell |\mathcal{B}|} \sum_{b \in \mathcal{B}} \mathbf{1}\{j \in b\}. \quad (55)$$

IID trajectories sample paired growth/output observations with these weights. Block trajectories sample starts uniformly and concatenate intact blocks, with an independent uniform initial phase. Each action index then has the same marginal law w . Chronological output marks refer to the ordinary call preceding the input difference; they have mean 817.6686 across all accepted transitions and 848.0400 under block weighting. The latter is the output invoice baseline for matched comparisons. Growth/output pooled correlation is .3882. The marked split θ is a separate retention/timing intervention, rather than an assumption that billed output exactly equals retained cold growth.

E.3 Reset and price parameters

The magnitude anchors $S = 65,588$ and $C = 4,382$ come from Lang-Watch’s reported medians in one practitioner’s 162-day study [11]. They are used jointly as a scenario, not as a jointly measured TraceLab mean. $B = 20,000$ is a within- S intervention. Increasing C to 20k preserves the remaining material by increasing S to 81,206. Adding stable material changes both S and B , whereas reclassification changes only B .

The quoted-vector catalog records USD/token and USD/million, processing tier, region, context/time band, effective write conventions, source URLs, and native CNY quotes separately. The Qwen Flash source discrepancy is recorded with the affected vectors. Constant input-price bands are projected independently, and provider context ceilings are not imposed on the replay.

Table 3 indexes the distinct evidence families before their detailed experiments.

F Versioned Working-State Experiments

F.1 Question, measured sizes and controlled task law

This experiment asks when immediate reconstruction is an adequate model of required working state. The engine separates current file versions, valid resident observations, qualifying read guards, physical text, snapshot recency and matching-prefix age. Obsolete text remains billed until removed; a retained fact can be semantically useful but computationally cold. These distinctions cannot be replaced by a single random reload size.

The pinned whole-file catalog comprises Jinja filters.py (13,876 tokens), Rich text.py (10,417), CPython asyncio/base_events.py (15,836) and Django query.py (21,189), a total of 61,318 under cl100k_base. Commits, byte counts and source hashes are in working-set-results.json. The catalog is a cross-repository size intervention, not a measurement that a typical issue requires

Table 3: Evidence and execution map for the distinct study families. Aggregate paths are relative to docs/research; runner paths to experiments. The symbolic task-efficiency row has no empirical success data. Shared source pools and measured/control distinctions are specified in the corresponding experimental appendices.

Study/estimand	Maintained aggregate	Rebuild runner
Versioned-file invoices and budgets	working-set-results.json	run_working_set_study.py; working_set_retention_sensitivity.py
Exact finite cells and continuum controls	paper-supplement-results.json	working_set_exact_control.py; structural_symbolic_analysis.py; continuum_threshold_analysis.py; export_paper_supplement.py
Renewal extraction and horizon confirmation	renewal-results.json	calibrate_renewal_trace.py; renewal_community_study.py
Price/mode/material contrasts	price-family-results.json; price-scenarios.json	run_price_sensitivity.py; price_family_crosscheck.py
Reachable-state control certificate	structural-results.json	structural_control_counterexample.py; summarize_structural_theory.py
Task-efficiency identities	Symbolic residuals in the generated paper cache	task_efficiency_derivation.py
Publication exhibits	Embedded standalone tables and plot coordinates	build_paper_figures.py; build_paper_exhibits.py

all four files. A 64-token wrapper per observation, 20k stable base and 4,382-token summary produce an eager rebuilt state of 85,956 tokens. The tokenizer is a reproducible proxy, not a Claude tokenizer.

Ordinary output uses 305,445 pinned TraceLab Claude observations, mean 840.3106, median 354 and p99 7,163 tokens; zeros remain. Length-16 output blocks have 250,166 eligible starts. Output-only resampling is intentional: the engine supplies file restoration and a controlled 256-token non-file append. Adding measured full-input growth would duplicate those inputs. The later renewal experiment uses full-input growth instead, and does not independently add this catalog. Absolute thresholds and costs from these two designs are not pooled.

The main workload has 400 actions, file requirement probability .85, .9 preference for a hot file, 25-action phases and .002 independent revision probability per file/action. Automatic observations are disabled. Request gaps are lognormal, mean 90 seconds and coefficient of variation 1.5; cache minimum is 1,024 tokens and lifetime 300 seconds. Ordinary uncached instructions have 128 tokens. The baseline price vector is (3, 3.75, .30, 15) USD/M; compactors use the same prices and summary size. Optional batched recovery inference generates 64 command tokens and uses 64 uncached instructions, while processing the whole existing context. Extra requests do not replace ordinary actions; no terminal compact is billed.

F.2 Restoration policies and category attribution

Table 4 separates eager and first-use restoration, retained observations with/without read guards, surviving stable prefixes and extra recovery inference. The coarse grid is 40–240k in 10k steps, with paired centered contrasts at $\pm 5k$. Discovery and independent confirmation each use 2,048 paths, seeds 20261006/20261007. Policies share an immutable ordinary task field, with separate output, gap and mutation streams. Discovery-selected choices are frozen for confirmation.

Eager warm minus first-use warm is \$7.635728, paired interval [7.592276, 7.679180]. Yet first use restores 801,484 payload tokens, more than eager’s 690,421; retention restores only 587,475 but costs more. The complete category attribution explains the ordering:

Policy	Input	Write	Read	Output
Eager warm	.4356	12.1026	12.1855	5.8786
First-use warm	.5145	9.0952	6.8057	6.5511
Retain + guard	.5064	9.7765	7.7633	6.3824

Entries are USD/task and reconcile to displayed rounding. Earlier cleanup pays more recurring summary output, but avoids larger prefix writes and repeated carries. Retaining payload avoids reads without avoiding its cold rewrite or subsequent processing. Losing guards can require a duplicate qualifying read even while text remains. Tool reread counts alone therefore cannot order the policies.

F.3 Mechanism matrix and feasibility

Table 5 changes one family of mechanisms at a time; both policies are separately selected on discovery under that intervention. Diffuse demand saturates the catalog sooner, raising first-use’s line and narrowing its benefit. Under mandatory demand, eager, first use and valid-guard retention have identical complete invoices: all must supply the same next input. Higher churn invalidates facts while stale text remains billed. Larger base size moves the physical floor without changing the primitive task law. Delaying compact/recovery by 360 seconds is a cache-expiry stress condition, not measured production latency.

Strict eager fails on every sampled trajectory at 40–80k: the required 85,956-token reset cannot fit. Strict first use fails at 40k. These points are infeasible, not inexpensive partial completions. No failed path is censored into a completed-task bill. At 80k, strict first use costs \$23.5539 versus atomic \$22.9665, through extra reset/recovery. In the 40-action case, eager selects 190k and only .0073 mean compactions, effectively no compaction, while first use selects 80k and 1.4165 mean compactions. Remaining work is important but does not imply one universal short-task policy.

F.4 Joint retention budget and paid recovery delay

Five budgets—0, 12k, 20k, 40k and 64k—are jointly selected with lines 60–160k in 10k steps. All preserve valid guards and the same live base; batched recovery inference is enabled. Discovery/confirmation use 1,024 paths each, seeds 20261010 and 20261011. Extra recovery delay is 0 or 360 seconds; compactor delay is zero, unlike the both-delayed matrix row.

Table 4: Phase-local policies at coarse discovery-selected lines. USD/task refers to all 400 actions; intervals are independent confirmation MC95 halfwidths. Reload is supplied payload excluding wrappers; warm is a surviving live 20k base boundary.

Policy	H (k)	USD/task	Reload (k)
Eager, cold base	130	31.2434 ± 0.0986	690.4
First use, cold base	90	24.2897 ± 0.0776	706.2
Eager, warm base	130	30.6022 ± 0.0963	690.4
First use, warm base	80	22.9665 ± 0.0731	801.5
Retain files and guards	90	24.4286 ± 0.0794	587.5
Retain files, lose guards	100	26.3346 ± 0.0817	727.6
Eager + recovery model	130	30.7680 ± 0.0967	691.4
First use + recovery model	80	23.7356 ± 0.0754	803.4

Table 5: Versioned-file mechanism interventions. E is eager warm and F first-use warm; thresholds are thousand tokens and bills USD/task. Only the short row has 40 actions; other rows have 400. The strict row rechecks the line after restoration for both policies.

Intervention	H_E (k)	USD $_E$	H_F (k)	USD $_F$
Phase-local	130	30.6022	80	22.9665
Diffuse demand	130	30.7454	120	29.2017
All files mandatory	130	30.8170	130	30.8170
Revision churn .02	130	32.2947	80	23.6580
Both extra delays 360s	130	31.8971	90	30.1490
Stable base 0	100	26.6669	60	18.9630
Stable base 40k	150	34.5201	100	26.9701
40 ordinary actions	190	2.9144	80	2.2108
Strict trigger	130	30.5878	80	23.5539

Table 6: Joint budget/threshold selection with model-mediated recovery. Subscripts denote recovery delay in seconds. USD/task intervals are independent confirmation MC95 halfwidths.

Budget (k)	H_0 (k)	USD $_0$	H_{360} (k)	USD $_{360}$
0	80	23.7657 ± 0.1021	90	31.4377 ± 0.1362
12	90	25.0624 ± 0.1104	100	31.7911 ± 0.1450
20	90	25.1175 ± 0.1094	100	31.9174 ± 0.1462
40	100	26.8237 ± 0.1150	110	32.3302 ± 0.1422
64	120	29.7522 ± 0.1344	120	31.4437 ± 0.1413

Zero retention wins discovery in both conditions. With no delay, retaining more is materially more expensive. With 360-second recovery delay, retaining the full catalog avoids many expired-cache requests and nearly ties dropping all files: zero minus 64k is $-.006029$, interval $[-.100050, .087993]$ USD. Intermediate retention still loses; zero minus 20k is $-.479665$, interval $[-.543733, -.415597]$. Partial retention can pay both the carry cost of kept files and the inference/expiry cost of missing ones. The nonmonotonic response is more informative than declaring the winning budget a universal retention rule.

F.5 Independent fine grids and finite-scale slopes

New discovery/confirmation samples use 4,096 paths each, seeds 20261008/20261009. Phase-local lines are 60–110k and mandatory lines 100–160k, both in 2k steps. Selected lines and sampled 1% regions are 78k, 68–86k and 126k, 116–138k. Confirmed bills are

\$22.954254, interval $[22.903439, 23.005068]$, and \$30.778354, interval $[30.711815, 30.844894]$. The paired centered secant is

$$\frac{\widehat{J}_N(H + \delta) - \widehat{J}_N(H - \delta)}{2\delta}, \quad \delta = 2,000. \quad (56)$$

Table 7 scales this to USD per extra 1k threshold tokens over the whole task. Its sign changes support distinct valleys. A grid minimum need not have zero centered secant: 78k has a small positive contrast. The sampled 1% regions describe loss basins rather than confidence sets for a globally unique optimum.

G Exact Finite Control and State Sufficiency

G.1 Exhaustive physical-state control

The same simulator is run on all $2^8 = 256$ ordinary demand paths. Its single file has 24 payload and two wrapper tokens, base/summary 32/8, ordinary retained input/output 20/10 and ordinary/compactor instructions 2/4. Demand is iid Bernoulli(α), $\alpha = 7/20$, on fixed

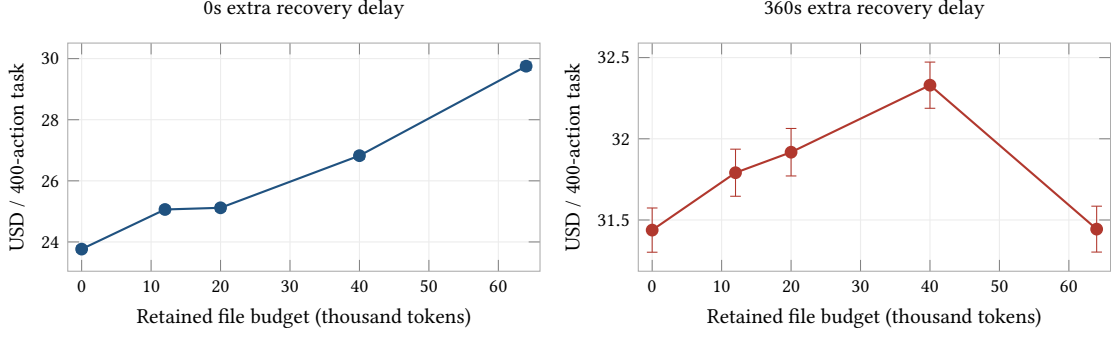


Figure 6: Independently confirmed retention-budget invoices after joint threshold selection. Bars are pointwise MC95 intervals. With zero recovery delay, extra retention costs more; with expiry-inducing recovery delay, full-catalog retention nearly ties dropping all files while intermediate budgets remain more expensive. The paired endpoint interval is reported in text, rather than inferred from overlap of unpaired error bars.

Table 7: Paired discovery fine-grid secants, USD per additional 1k threshold tokens over a 400-action task.

Law	H (k)	USD/1k	MC95 interval
Phase-local	68	-0.05823	$[-0.06095, -0.05552]$
Phase-local	74	-0.01399	$[-0.01685, -0.01112]$
Phase-local	78	+0.00341	$[+0.00036, +0.00645]$
Phase-local	86	+0.02820	$[+0.02484, +0.03156]$
Mandatory	116	-0.06244	$[-0.06497, -0.05991]$
Mandatory	126	-0.00114	$[-0.00411, +0.00183]$
Mandatory	138	+0.03550	$[+0.03169, +0.03931]$

ordinary-action indices. There is no expiry or mutation, initial base is warm, cache minimum is 16 and triggers are atomic. This small fixture is an exact extraction basis, not an engineering-scale size estimate.

For path b with $k(b)$ demands,

$$J_N(\pi, H) = \sum_{b \in \{0,1\}^8} \alpha^{k(b)} (1 - \alpha)^{8-k(b)} \mathbf{p}^\top \boldsymbol{\ell}_{\pi, H}(b). \quad (57)$$

Every integer ledger is extracted from the actual engine before expectation or symbolic lifting. Every path completes all eight actions; there is no terminal compact. All lines 1–299 are scanned and the constant infinite tail starts at 269. Read stress changes only p_r from .30 to 3 USD/M.

Default first use costs \$.002553517622 without compaction. Read stress selects 157–178 at \$.004964155380, versus \$.005691137297 without compaction, a 12.77% reduction. Default has five upward and sixteen downward nonzero jumps; stress has nine and twelve. Default cost even rises from \$.002791064468 at 178 to \$.002797940778 at 179. Cycle regrouping and the unfinished last cycle matter. Because $x \geq H$ is equivalent to $x \geq \lceil H \rceil$ for integer x , induction over the identical event sequence proves $J_N(\pi, H) = J_N(\pi, \lceil H \rceil)$ when the line enters only these comparisons. The real stress optimum is (156, 178]. A zero ordinary derivative between cells is not an optimization answer; signed jumps contain the decision-relevant differences.

G.2 Endogenous first use and batching reversal

At $H = 100$, actual-engine prefix reruns identify first-cycle action count T and first required-use index K_f :

$$\mathbb{E}T = 2.4225, \quad \mathbb{P}(K_f \leq T) = .725375, \quad \mathbb{E}(T - K_f)_+ = .35. \quad (58)$$

The predictable carry identity $\mathbb{E}T - \mathbb{P}(K_f \leq T)/\alpha = .35$ is exact. The independent-clock shortcut $1 - \mathbb{E}(1 - \alpha)^T = .6399771875$ is lower by .0853978125. Restoring the observation changes crossing, so the stopping clock’s marginal distribution cannot determine the joint restoration law.

A deterministic two-file control uses eight actions, two 20-token files, base/summary 32/8, and an 80-token background append on action 1. Requirements are a on action 1, b on 2 and a on 3, then none; $H = 120$ produces one reset. Tool-only recovery costs \$.0010422 first use versus \$.0011292 eager. Adding a model recovery request with 80 instruction and eight output tokens per batch reverses the ordering: \$.0022830 versus \$.0019476. Eager supplies 80 file tokens versus first use’s 60 but makes two recovery rounds rather than three, saving whole-context processing and later command carries. Its bill is 14.69% lower. Independent per-file fees cannot express that batching benefit.

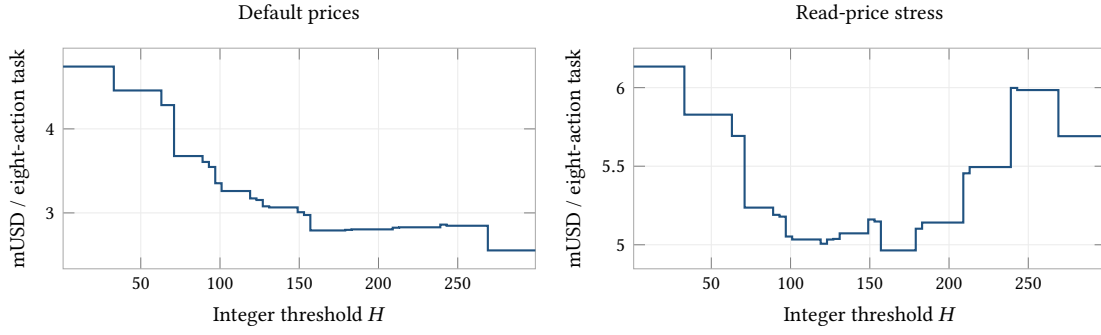


Figure 7: Exhaustive eight-action first-use invoices at integer configurations. Default prices favor no compaction; read-price stress favors the entire 157–178 interval. Multiple upward/downward jumps exclude an automatic unimodal-search assumption.

Table 8: Exact minimizing configurations over six fixed policies. Bills are milli-USD per eight-action task, including the infinite no-compact tail.

Policy	Default H	mUSD	Read-stress H	mUSD
First use	≥ 269	2.553518	157–178	4.964155
Eager	≥ 269	2.570400	157–178	5.196000
Retain, lose guard	≥ 269	2.553518	179–182	5.326293
Retain + guard	≥ 269	2.553518	157–178	5.071671
Stable first use	≥ 269	2.553518	157–178	4.940155
Stable retain + guard	≥ 269	2.553518	157–178	5.047671

G.3 Symbolic demand response and global price faces

The exact first-use occupation polynomial at $H = 100$ is

$$m_i(\alpha) = 58, \quad m_o(\alpha) = 104, \quad (59)$$

$$m_w(\alpha) = 320 + 208\alpha - 78\alpha^2 - 52\alpha^3 + 26\alpha^4, \quad (60)$$

$$m_r(\alpha) = 578 + 156\alpha - 26\alpha^2. \quad (61)$$

At baseline prices,

$$J_{100}(\alpha) = \frac{3}{10^7} (325\alpha^4 - 650\alpha^3 - 1001\alpha^2 + 2756\alpha + 10358). \quad (62)$$

Its derivative is $39(50\alpha^3 - 75\alpha^2 - 77\alpha + 106)/(5 \cdot 10^6)$, positive on $[0, 1]$, and the branch is concave in demand. But at $H = 71$, $J'_{71}(7/20) = 1764843003/320000000000 > 0$. Price-envelope concavity does not imply demand concavity.

For any fixed finite policy, $m_j(\alpha) = \sum_b \ell_j(b) \alpha^{k(b)} (1 - \alpha)^{N-k(b)}$ has degree at most N . Grouping by demand count gives a Bernstein basis with conditional mean occupations as coefficients. Differentiating the finite sum for $0 < \alpha < 1$ yields

$$\partial_\alpha m_j = \text{Cov}(\ell_j, K_d) / (\alpha(1 - \alpha)), \quad (63)$$

where K_d counts all ordinary-index demands. This preserves covariance between demand and changed schedules. At endpoints use the polynomial derivative, not division by zero.

At 100, stable-prefix survival changes cost by $96(p_r - p_w)$ independently of demand. Retain-with-guard minus first use is exactly

$$26\alpha\bar{\alpha}\{p_w P_w(\bar{\alpha}) + p_r P_r(\bar{\alpha})\}, \quad (64)$$

where $\bar{\alpha} = 1 - \alpha$, $P_w(t) = t^6 + 2t^5 + 2t^4 + 3t^3 + 4t^2 + 3t$ and $P_r(t) = t^5 + 2t^4 + 3t^3 + 4t^2 + 5t + 3$. Nonnegative coefficients certify domination of this retention branch throughout $0 < \alpha < 1$ for positive write/read rates. At 7/20, reloads fall from 2.361756 to .968136 while cost rises \$.000130144444.

The nine regions use 2,196 exact endpoint/tail inequalities. Minimizing price over the occupation convex hull exposes its lower faces; an active policy region intersects comparisons against every candidate. A pairwise crossover alone cannot certify global optimality.

G.4 Reachable states and an exact omitted-information gap

Let x, w be retained length/live prefix, b_0, c_s reset base/summary, k_c, e compactor/ordinary instructions, b_a, o_a ordinary input/output, and r_f, d_f, α_f valid-file indicator, payload and unrevealed demand probability. Compact discards the fact and leaves reset material cold:

$$\mathbf{m}^{\text{continue}} = (e, x - w + b_a + \alpha_f d_f (1 - r_f), w, o_a), \quad (65)$$

$$\mathbf{m}^{\text{compact}} = (x - w + k_c + e, b_0 + c_s + b_a + \alpha_f d_f, w, c_s + o_a). \quad (66)$$

The exact compact disadvantage is

$$\Delta(r_f) = p_i k_c + (p_i - p_w)(x - w) + p_w(b_0 + c_s) + p_o c_s + p_w \alpha_f d_f r_f. \quad (67)$$

Read billing cancels because both actions read old w once; it is not omitted. If the first four terms lie between $-p_w \alpha_f d_f$ and zero, absent and valid states have opposite unique controls, an open parameter region.

Table 9: Certified global faces of all 122 extracted policy/cell lines. Demand is $7/20$, $p_i/p_w = 4/5$, $p_o/p_w = 4$. Interiors list active policies; boundaries may tie neighboring faces. First-interval no-compaction policies tie. This certifies the extracted family, not all possible controllers.

p_r/p_w lower	Upper	Active policy	Integer H
0.000000	0.184251	first use	≥ 269
0.184251	0.833791	stable first use	157–178
0.833791	1.000000	stable first use	119–122
1.000000	1.044783	first use	119–122
1.044783	1.661480	first use	97–100
1.661480	2.164030	first use	89–92
2.164030	2.982955	first use	71–88
2.982955	3.327273	first use	63–70
3.327273	∞	first use	33–62

With all four categories positive, set $x = 10,000$, $w = b_0 = 1,000$, $c_s = 100$, $d_f = 1,000$, $\alpha_f = 1/2$, $k_c = e = 128$ and $b_a = o_a = 2$. Valid continue costs \$.0344715, absent continue \$.0363465 and compact \$.0356055 in both. The states are reached from a warm 1k base by 8k output plus a qualifying 1k file observation, or 9k output without it. The next demand remains hidden until after the decision; first invoices are sunk. The cold-prefix variant sets $w = b_0 = 0$, $d_f = 4,000$ and starts with equal-invoice histories: 6k background plus 4k file, or 10k background. A gap beyond TTL makes both cold, giving Discussion’s bills.

If valid-state probability is ω and a length-only controller compacts with probability ρ , its state-aware regret is

$$\omega\rho\Delta(1) + (1-\omega)(1-\rho)(-\Delta(0)). \quad (68)$$

The minimum is $\min\{\omega\Delta(1), (1-\omega)(-\Delta(0))\} > 0$ for $0 < \omega < 1$. Randomization cannot repair omitted information. At equal weights the cold gap is \$.0011295 and the all-four-active gap \$.0003705.

G.5 Bellman/flow certificate and sampling crosscheck

For a finite observable sufficient state, a macroaction includes paid reconstruction and one ordinary action. Expected usage $\mathbf{m}_n(s, a)$ and controlled kernel P_n give

$$V_N = 0, \quad V_n(s) = \min_a \{ \mathbf{p}^\top \mathbf{m}_n(s, a) + \sum_{s'} P_n(s'|s, a) V_{n+1}(s') \}. \quad (69)$$

Usage and next state may be correlated. Sufficiency requires identical conditional joint usage/transition laws for histories mapped to the same state under every admissible action. Length alone fails here.

For the equal-weight one-stage cold states, occupations are valid-continue, valid-compact, absent-continue and absent-compact. Flow constraints are $F\mu = (1/2, 1/2)^\top$, $\mu \geq 0$, with

$$F = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}. \quad (70)$$

The invoice vector is

$$\mathbf{c} = \frac{1}{2000000} (75843, 80361, 90843, 80361). \quad (71)$$

Primal occupation is $(1/2, 0, 0, 1/2)$. The dual potentials and value are

$$\mathbf{v} = (75843, 80361)/2000000, \quad V = 39051/1000000. \quad (72)$$

Primal and dual agree. The nonnegative slacks, multiplied by one million, have entries 0, 2259, 5241 and 0. This certifies state-aware control for the declared one-stage action set, not an unrestricted compressor or the earlier history-generation decisions.

A 16,384-path control at seed 20261006 gives first-use bill \$.003352176 with SE $8.2974 \cdot 10^{-7}$ USD, versus exact \$.003353096. Retain-with-guard minus first use is \$.000129660, paired SE $7.2593 \cdot 10^{-7}$, versus exact \$.000130144. Stable-prefix saving is \$.0003312 on every path. These checks connect exact algebra, implemented stopping mechanics and sampling without claiming the fixture estimates production demand.

H Continuous Sensitivity and Scale Separation

H.1 An exact continuously uncertain task budget

For the exact finite invoice, draw $Z \sim N(0, 1)$ once independently per task and keep $H = \max(1, h + \sigma Z)$ fixed throughout that task. This models uncertainty in a whole-task budget, not a fresh jitter at each request. With initial cell bill J_0 and jump Δ_b at boundary b ,

$$J_\sigma(h) = J_0 + \sum_b \Delta_b \Phi((h-b)/\sigma), \quad (73)$$

$$J'_\sigma(h) = \sum_b \Delta_b \phi_\sigma(h-b), \quad (74)$$

$$J''_\sigma(h) = - \sum_b \Delta_b (h-b) \phi_\sigma(h-b)/\sigma^2. \quad (75)$$

These continuous derivatives preserve endogenous file, cache and reset mechanics through the exact cell bills. SymPy verifies them with demand and prices symbolic. Independent normal-probability integration over original cells agrees to within $2 \cdot 10^{-17}$ USD on the evaluation grid.

The deterministic read-stress minimum occupies [156, 178]. At $\sigma = .25$, curvature near 10^{-421} and a local bias bound near $10^{-425.94}$ USD make a mathematically isolated point practically unidentifiable. At $\sigma = 1$, the bias bound is $6.0883 \cdot 10^{-32}$. Wider kernels mix remote jumps and move the valley nonmonotonically. Roots are detected on a .05-token grid and refined with Brent’s method; the search

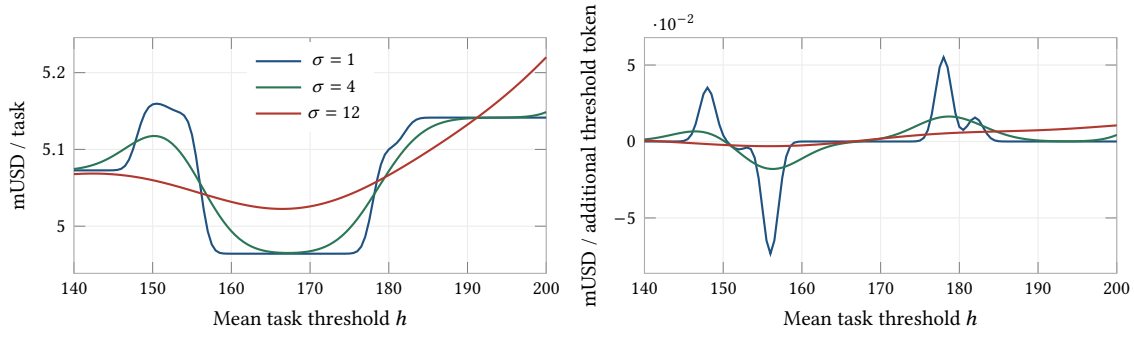


Figure 8: Exact whole-task uncertain-budget response of the read-stress fixture. Left: expected bill at three declared uncertainty scales; right: derivatives of those expectations. Curves evaluate the executed exact jump law, rather than fitting production threshold uncertainty.

Table 10: Detected local valleys of the uncertain-budget controller. σ is in fixture token units. Below-resolution bias is not a mathematical zero; logged curvature preserves tiny nonzero sensitivity.

σ	Valley mean	USD	Bias USD	$\log_{10} J'' $
0.25	167.000815	0.004964155380	Below resolution	-421.45
1	167.013037	0.004964155380	Below resolution	-29.13
4	167.201916	0.004965108998	$9.5362e - 07$	-6.30
12	166.453451	0.005022490446	$5.8335e - 05$	-6.30

characterizes local features rather than certifying all roots. For default prices the global answer remains the no-compaction tail. Its local valley moves from 167.0093 at $\sigma = .25$ to 183.5670 at $\sigma = 12$, but the latter bill \$.002801703928 exceeds the \$.002553517622 tail. A mixture cannot beat the best original policy; a lower smoothed bill at one mean is not a globally better policy.

H.2 Bias, gradient convergence and useful approximation

Away from jumps, normal tails give

$$|J_\sigma(h) - J(h)| \leq \sum_b |\Delta_b| \Phi(-|h - b|/\sigma). \quad (76)$$

At a nonvanishing jump, smoothing tends to the average of neighboring values, leaving half-jump endpoint bias. Large reconstruction transitions must remain visible rather than be relabeled microscopic noise. For a different question, suppose a finite family and valid smooth extension satisfy $\|J_\delta - J_c\|_\infty \leq \varepsilon_\delta$ and J'_c is L_1 -Lipschitz. Then

$$\|J'_{\delta,\sigma} - J'_c\|_\infty \leq \sqrt{2/\pi}(\varepsilon_\delta/\sigma + L_1\sigma). \quad (77)$$

Differentiate the cost-error convolution and use $\|\phi'_\sigma\|_1 = \sqrt{2/\pi}/\sigma$ for the first term; bound the expected change in J'_c by $L_1\mathbb{E}|\sigma Z|$ for the second. Positive constants balance at $\sigma = \sqrt{\varepsilon_\delta/L_1}$, giving error $2\sqrt{2/\pi}\sqrt{\varepsilon_\delta L_1}$. Boundaries require that valid extension or explicit tail remainders. This is a whole-objective approximation theorem, not a diffusion assumption about tool-driven jumps. If an objective surrogate uniformly errs by ε and its selected policy has surrogate regret ζ , the actual regret is at most $2\varepsilon + \zeta$, by comparing both

selected and true-optimal costs to their surrogates. That is a useful control guarantee without asserting microscopic smoothness.

H.3 Executed finite-to-continuous law example

For U uniform on $[0, 1]$, one-event cost $1\{U \geq h\}$ has interior expectation $1 - h$ and derivative -1 . Replace it by m finite equiprobable midpoints u_k , $\delta = 1/m$. Cost error is at most $\delta/2$, while its signed derivative has midpoint atoms of weight $-\delta$. With one clipped normal task threshold,

$$J'_{\delta,\sigma} = -\delta \sum_k \phi_\sigma(h - u_k), \quad J'_{c,\sigma} = -\{\Phi(h/\sigma) - \Phi((h-1)/\sigma)\}. \quad (78)$$

This is a separate analytic approximation example, not an empirical law for the file engine. Table 11 compares $\sigma = \sqrt{\delta}$ with $\sigma = \delta/8$ at a gap and neighboring midpoint near $1/2$.

Coarse slopes approach -1 ; fine gap/midpoint slopes remain near -0.0214 and -3.19154 . Cost convergence alone does not imply derivative convergence. Integration by parts against the Gaussian kernel and the $\delta/2$ CDF error give $|J'_{\delta,\sigma} - J'_{c,\sigma}| \leq \delta/(\sigma\sqrt{2\pi})$. This justifies $\sqrt{\delta}$ for this example once interior boundary tails vanish. Its sharper scalar bound and the general objective bound answer distinct questions. Finite realizations permit useful continuous laws; the decision-relevant scale and approximation error determine when.

Table 11: Finite midpoint laws approaching a continuous uniform law. Coarse bandwidth is $\sqrt{\delta}$; fine bandwidth $\delta/8$. Cost converges in both cases, but fine slopes retain lattice-scale peaks.

m	Cost bound	Coarse slope	Fine gap slope	Fine midpoint slope
16	0.03125000	-0.95506110	-0.00214128	-3.19153824
64	0.00781250	-0.99993735	-0.00214128	-3.19153824
256	0.00195312	-1.00000000	-0.00214128	-3.19153824
1024	0.00048828	-1.00000000	-0.00214128	-3.19153824

I Renewal Laws, Approximation and Horizon Experiments

I.1 Distributional comparisons rather than a named prior

Exponential growth of mean g has $U = 1 + L/g$ and $D = L + L^2/(2g)$, giving $L_* = \sqrt{g^2 + 2gA/c} - g$. The initial renewal mass is the first ordinary request, not an optional correction. The fluid gap $\sqrt{2gA/c}$ instead relaxes the stopping clock. For Erlang- k increments, write $\omega_j = e^{2\pi i j/k}$ and $r_j = (k/g)(\omega_j - 1)$. Laplace residues give

$$U_k = L/g + (k+1)/(2k) + \sum_{j=1}^{k-1} \frac{\omega_j e^{r_j L}}{k(\omega_j - 1)}, \quad (79)$$

$$D_k = L^2/(2g) + (k+1)L/(2k) + \sum_{j=1}^{k-1} \frac{\omega_j (e^{r_j L} - 1)}{k(\omega_j - 1)r_j}. \quad (80)$$

Conjugate poles make the sums real; for $k = 2$, $D_2 = L^2/(2g) + 3L/4 + g(1 - e^{-4L/g})/16$. Symbolic Laplace checks verify $D' = U$ and the rate derivative. Bracketed inversion is more useful than forcing an elementary inverse. Two-exponential mixtures provide an analytic high-dispersion alternative; moment matching is checked against the empirical crossing, not assumed accurate merely because a distribution can be fitted.

Deterministic increments have $T = \lceil L/g \rceil$ and cell-dependent rate $A/m + cg(m-1)/2$, ignoring constant baseline terms. An optimal cell satisfies $g(m-1)m/2 \leq A/c \leq gm(m+1)/2$. For $t = L/g - \lfloor L/g \rfloor$, $D = L^2/(2g) + L/2 + gt(1-t)/2$ has a genuinely periodic lattice ripple. For a nonarithmetic law with finite third moment, the regular renewal expansion instead gives

$$D(L) = \frac{L^2}{2g} + \frac{\mathbb{E}G^2}{2g^2}L + \frac{(\mathbb{E}G^2)^2}{4g^3} - \frac{\mathbb{E}G^3}{6g^2} + o(1). \quad (81)$$

The constant follows from the renewal remainder, not just series matching. Write $m_k = \mathbb{E}G^k$, $a = m_2/(2g^2)$, $C_0 = m_2^2/(4g^3) - m_3/(6g^2)$ and $P(L) = L^2/(2g) + aL + C_0$. The equation $D = L + F * D$ gives $q = r + F * q$ for $q = D - P$, with $r(L) = -\mathbb{E}[P(L - G)\mathbf{1}\{G > L\}]$. Finite m_3 makes this linear combination of integrated tail powers and tail probability directly Riemann integrable. Termwise Tonelli gives

$$\int_0^\infty r(L) dL = -m_3/(6g) + am_2/2 - C_0g = 0. \quad (82)$$

The nonarithmetic key renewal theorem gives $q \rightarrow 0$, establishing the constant as well as leading terms. The empirical pool's large jumps dominate that moment, motivating finite-scale comparison in Table 12.

Table 12: Baseline selections separating growth-law approximation from marked accounting. All use the same mean, reset anchors and prices.

Method/accounting	H (k)
Fluid mean-growth relaxation	115.428
Exponential, matched mean, core	113.918
Hyperexponential, two moments, core	112.909
Second-moment asymptotic, core	112.780
Third-moment asymptotic, core	113.511
Empirical renewal, core	113.190
Empirical renewal, marked	112.848

The empirical core crossing is 113.190k and marked line 112.848k; fluid gives 115.428k, second-moment asymptotic 112.780k and third-moment 113.511k. Approximation order alone does not rank finite-scale fidelity. The finite empirical law is arithmetic. For maximal span δ and phase $t = L/\delta - \lfloor L/\delta \rfloor$, its finite-support expansion uses $C_\delta = C_0 - \delta^2/(12g)$ and adds ripple $\delta^2 t(1-t)/(2g)$. Expand $\delta z / \{(1-z)^2(1 - \mathbb{E}z^{G/\delta})\}$ at $z = 1$; other finite-support poles are outside the unit circle when positive support indices have gcd one. The ripple is at most $\delta^2/(8g)$. The recorded third-moment prediction uses that span correction, rather than transplanting an unmodified nonarithmetic constant to token counts. Core versus marked differences change accounting; exponential versus empirical differences change the law. Their proximity must also be read against the broad loss basin rather than point-estimate precision alone.

I.2 Crossing marks and numerical refinement

Table 13 separates cheap analytic probes from community estimation. At small $A/(cg) = .2$, moment inversion under the hyperexponential law errs by over 600%: a large negative asymptotic constant is not a valid description of the near-zero renewal region. At ratio 500 the same approximation is accurate to a tiny relative error. Erlang phase laws display smaller but still visible transient errors at small ratios. All moments existing is not enough to guarantee finite-scale asymptotic accuracy; the crossing must be compared to the transient renewal law. The exponential quadratic is exact at every ratio because its renewal function has no transient remainder.

Table 14 records selected cycle duration and terminal increment for each intervention. The baseline terminal increment is about 4,733 tokens, well above the ordinary mean 1,533: substituting an unconditional mark loses stopping-boundary selection. Lower reliability raises carry price and shortens cycles; larger summary changes output and the floor. Setting $\theta = 0$ changes billing but not the core root.

Table 13: Executed dimensionless closed-law controls, not fitted community workloads. Exact gaps invert the analytic renewal law; fluid ignores crossing correction; moment error compares the quadratic third-moment root to the exact gap. Hyperexponential CV is two.

Law	$A/(cg)$	CV^2	Exact L/g	Fluid L/g	Moment error (%)
exponential	0.2	1.000	0.183216	0.632456	-0.0000
erlang2	0.2	0.500	0.195838	0.632456	-15.6699
erlang3	0.2	0.333	0.198762	0.632456	-15.5891
erlang8	0.2	0.125	0.199993	0.632456	-9.6478
hyperexp cv2	0.2	4.000	0.175557	0.632456	+618.6564
exponential	2	1.000	1.236068	2.000000	+0.0000
erlang2	2	0.500	1.356668	2.000000	-0.0096
erlang3	2	0.333	1.406029	2.000000	+0.0040
erlang8	2	0.125	1.474842	2.000000	+0.0263
hyperexp cv2	2	4.000	1.092340	2.000000	+56.8262
exponential	500	1.000	30.638584	31.622777	+0.0000
erlang2	500	0.500	30.879693	31.622777	+0.0000
erlang3	500	0.333	30.960794	31.622777	-0.0000
erlang8	500	0.125	31.062685	31.622777	-0.0000
hyperexp cv2	500	4.000	29.339440	31.622777	+0.0000

Table 14: Renewal interventions and crossing diagnostics: U is mean actions per cycle and $z = \mathbb{E}G_T$ terminal growth. Summary 20k raises S to 81,206 to preserve other material.

Intervention	H (k)	USD/action	U	z (tokens)
Baseline	112.848	0.052701	32.42	4733.0
No stable base	119.498	0.054696	36.76	4763.6
$q = .8$	91.003	0.108699	18.12	4571.1
$q = .5$	83.093	0.186518	12.93	4428.9
Summary 20k	152.666	0.064648	48.23	4828.1
$\theta = 0$	113.193	0.053270	32.64	4734.7

Refinement from 25 to 5 tokens moves baseline H by 15 and rate by $4.1128 \cdot 10^{-7}$ USD/action. Convolution residual is near roundoff. At 150k gap the tail lower envelope .060846 exceeds the .052701 minimum, using core monotonicity and finite-support maximum crossing size, not universal marked unimodality. Each price case records its own positive tail margin; raw finite trajectories remain unbinned.

I.3 Dependence and complete horizon confirmation

The corrected block design matches observation-appearance marginals and randomizes initial phase. Before correction, block-position means ranged from 1,655.13 to 1,470.58, confounding correlation with finite-start weighting. With matched marginals, length-16 growth-sum variance is 3.1555 times IID; 4,000-action invoice-rate variance is about 4.14 times larger under blocks. The mean basin changes little. An all-accepted IID control selects about 116k, but its weighting differs; that is not a correlation effect.

At 32 actions, IID/block selections 150.588k/165.588k independently save $.002269 \pm .000268$ and $.001824 \pm .000234$ USD/action, or 4.78%/3.94%. At 400 and 4,000, local refinements have unresolved or zero gains. The warm initial state is sunk, and no terminal compact is charged, so short tasks have less time to repay reconstruction. The unconstrained no-compact benchmark is in the short sampled

1% region but not the long one. A binding provider capacity ceiling removes that alternative rather than making its unconstrained bill actionable.

J Price Sensitivity and Material Interventions

J.1 Differential predictions versus finite contrasts

Table 16 separates core gap elasticities, optimized invoice shares and full marked selections after a finite 20% price change. Core gap elasticities sum to zero, expressing common-scaling invariance; bill shares sum to one, expressing degree-one homogeneity. Core physical- H elasticities are gap elasticities times L/H . At perfect hits, uncached input affects setup mainly through 128 compactor instruction tokens. Input $\times 1.2$ shifts core H by about eight tokens but marked H by 285: the terminal correction changes attribution even when its baseline displacement is modest. Ordinary output is an action-indexed baseline when growth is fixed; recurring summary output instead changes reset cost.

Table 17 includes both directions of all four price changes, sign controls and material/timing paths. Analytical old-policy excess divides by the new optimum, whereas block differences compare complete returned and reference bills. Price-only groups reuse immutable four-category ledgers; mechanism changes retain common

Table 15: All independent horizon contrasts. Δ is selected policy minus the 112.848k analytic policy, in 10^{-3} USD/action with paired MC95 halfwidths. Choices are frozen on separate discovery samples.

Law	N	H (k)	USD/action	$10^3 \Delta$ USD/action
IID	32	150.588	0.045231	-2.26855 ± 0.26803
IID	400	111.848	0.052310	$+0.01595 \pm 0.03242$
IID	4000	113.848	0.052695	$+0.01026 \pm 0.01155$
Blocks	32	165.588	0.044524	-1.82448 ± 0.23393
Blocks	400	112.848	0.052235	$+0.00000 \pm 0.00000$
Blocks	4000	113.848	0.052671	$+0.00940 \pm 0.01375$

Table 16: Baseline differential and finite price sensitivity. Core gap responses and optimized bill shares describe different quantities; the last column uses the full empirical marked ledger after one-price +20%.

Price	Core gap elasticity	Bill elasticity	$H_{+20\%}$ (k)
Input	+0.000824	0.015823	113.133
Write	+0.366752	0.198757	115.923
Read	-0.508587	0.505569	108.668
Output	+0.141011	0.279850	114.183

action-index fields. Confirmation uses 512 paths of 4,000 actions per mode. Price-sign controls use the unchanged-price optimum of their own mechanism, not a global 113k reference that would confound cache/reset and price effects.

J.2 Reversal mechanisms and their monetary size

Write $\times 1.2$ at $q = 1$ moves 112.848k to 115.923k; at $q = .8$ it moves 91.003k to 90.403k. The core infinitesimal write boundary is .8857444, while the marked empirical 5% finite contrast crosses near .90. The latter is an atomic finite switch, not another estimate of the same differential object. The thin-volatile control holds $S = 65,588$, $B = 64,000$, $C = 100$ and $q = 1$. With $\theta = 1$, write $\times 1.2$ moves 79.368k down to 78.893k; with $\theta = 0$, it moves 80.248k up to 80.603k. This isolates the cold terminal mark priced at p_i before a normal request can write it at p_w . A thin stable-heavy control at $q = .5$ increases read fourfold and moves 76.703k to 76.883k: recurring stable compactor reads can dominate ordinary history carry. Valid reversal does not imply material retuning value: old-policy excess is .0061%, .0052% and .0015% for these miss-write, cold-tail-write and read controls.

J.3 Material paths and locally valuable price directions

Reclassifying existing 20k fixes S and increases B ; adding stable 20k increases both. At baseline their core physical responses per token are approximately $-.352$ and 1.031 . Added volatile material gives 1.383 and extra summary preserving other material gives 2.915 . The marked thresholds and block gains are in Table 17. Scaling the whole retained-growth law by two, leaving billed output unchanged, has core gap-scale elasticity .479 and marked $H = 130.803$ k. Each intervention specifies a direction in parameter space; “more context” alone cannot define a derivative.

At a smooth core root, $\mathbf{w} = \mathbf{a} - R\mathbf{v}$ is the one locally valuable price direction. Orthogonal perturbations do not move the line to first order, and $\mathbf{w} \cdot \mathbf{p} = A - Rc = 0$ puts common scaling in the null space. Appendix D derives rank-one optimized curvature and quadratic retuning loss. At atoms, exact price-linear margins replace smooth curvature. This explains how visibly moving lines can coexist with tiny gains inside a broad basin.

J.4 Lifetime tariff and reliability break-even

With short $q = .8$, long tariff needs $q = .896281$ at the old line, or .895035 after retuning. At long $q = .95$, paired total expense changes by $-.0169031452 \pm .0001051715$ USD/action, about 23.3% cheaper. Changing only tariff and keeping $q = .8$ adds $.0278259834 \pm .0000634353$ USD/action. At short $q = .98$, even perfect long reuse misses break-even by only about .107%, a small conditional margin. Required improvements are not observed causal gains from an advertised cache lifetime; they specify what a useful timing measurement would need to establish.

K Complete Quoted-Vector Experiments

K.1 Controlled comparison and all model configurations

The 2026-10-06 catalog has 38 official USD/M four-vectors: 17 OpenAI entries over 12 GPT models, 12 Qwen, four DeepSeek and five Claude. Effective write equals miss input when there is no separate creation fee; input and write categories remain disjoint. Specific Qwen creation quotes are used with the conflict against generic multiplier prose recorded. Native CNY quotes are stored separately, never converted by an invented exchange rate. Every vector holds $S = 65,588$, $B = 20,000$, $C = 4,382$, $K = I = 128$, $\theta = 1$ and the same growth/output cohort. Three hit conditions per vector plus 29 controls yield 143 analytic cases. Each marked binned-law minimum has a right-tail bound. IID/block confirmation each uses 512 paths of 4,000 actions, repricing shared ledgers. A different model

Table 17: Executed price and material controls. Sign-test references are same-mechanism old-price optima; material/growth references are the original baseline. Block Δ is retuned minus reference expense, in 10^{-3} USD/action with paired MC95 halfwidths. Small unresolved differences are retained rather than selecting only significant shifts.

Intervention	Old H (k)	New H (k)	Excess (%)	Block $10^3 \Delta$
Input $\times .8$	112.848	112.548	0.0005	-0.00655 ± 0.01002
Input $\times 1.2$	112.848	113.133	0.0005	-0.00258 ± 0.00984
Write $\times .8$	112.848	109.568	0.0643	-0.04190 ± 0.01332
Write $\times 1.2$	112.848	115.923	0.0522	-0.02937 ± 0.01547
Read $\times .8$	112.848	118.528	0.1652	-0.08547 ± 0.01257
Read $\times 1.2$	112.848	108.668	0.1095	-0.06842 ± 0.01639
Output $\times .8$	112.848	111.478	0.0114	-0.00651 ± 0.01399
Output $\times 1.2$	112.848	114.183	0.0097	-0.01539 ± 0.01433
$q = .8$, write $\times 1.2$	91.003	90.403	0.0061	$+0.00750 \pm 0.02986$
Thin volatile, write $\times 1.2$	79.368	78.893	0.0052	-0.00309 ± 0.00310
Thin, $\theta = 0$, write $\times 1.2$	80.248	80.603	0.0026	-0.00158 ± 0.00221
Thin, $q = .5$, read $\times 4$	76.703	76.883	0.0015	$+0.00875 \pm 0.02253$
Reclassify stable 20k	112.848	105.158	0.3551	-0.17697 ± 0.01290
Add stable 20k	112.848	133.463	3.6689	-2.16364 ± 0.02053
Add volatile 20k	112.848	140.038	6.1740	-3.77233 ± 0.02811
Summary 20k, preserve material	112.848	152.666	10.8420	-7.02670 ± 0.04726
All growth pre-call	112.848	113.193	0.0007	$+0.00054 \pm 0.01055$
Whole growth law $\times 2$	112.848	130.803	1.4662	-0.96840 ± 0.02043

Table 18: Required reliability for Sonnet 5.5’s higher-write one-hour tariff to match the short tariff under otherwise identical conditions. Dashes mean no $q \leq 1$ breaks even.

Short q	Required q , old line	Required q , retuned
0.50	0.715161	0.714243
0.80	0.896281	0.895035
0.90	0.955279	0.953985
0.95	0.984268	0.982977
0.98	–	–
1.00	–	–

label does not change behavior, tokenizer, success or capacity in this projection.

GPT-6 Astra/Sol/Luna and GPT-5.6 Sol short vectors normalize to (1, 1.25, .10, 5), selecting 112.848k at $q = 1$. GPT-6.1 Sol lowers relative read to .05 and selects 132.838k; GPT-4.1 has .25 and selects 92.403k. Some long quotes double input/write/read but multiply output by 1.5, changing relative summary cost rather than uniformly scaling. A line below a provider’s long-price boundary does not mean that long quote is actually incurred. DeepSeek peak is exactly twice off-peak, so constant-band lines agree at every q . Flash’s .003 USD/M reads select 162.003k at perfect reuse, 116.923k at .95 and 94.423k at .8. Perfect-hit block retuning savings are about 5.585% Flash, 1.566% GPT-6.1 Sol and 3.591% GPT-4.1. Twenty-seven proportional-vector controls check identical selections and proportional analytic/simulated bills; all 35 earlier analytic cases remain exactly unchanged in the expansion.

K.2 Cache modes, domination and request bands

Singapore Max’s explicit write/read is 2.5/.17 versus implicit 2/.25 USD/M. At .8 the premium wins; at .95 and 1 the repeated-read discount wins. Flash’s selected specific table instead keeps input/read/output unchanged and raises explicit writes .15 to .20. At fixed feasible policies and equal q , explicit is componentwise

dominated: every bill is at least the implicit bill, hence so is the optimized bill. Observed conditional excesses are 20.53%, 12.39% and 6.30%. Actual modes can change reliability, a joint mechanism distinct from that price comparison. Plus long quotes uniformly scale short by three within each mode, so independently projected lines agree.

For request bands b , exact accounting is

$$J(\pi) = \sum_b p_b^\top m_b(\pi), \quad (83)$$

including ordinary, compactor and recovery requests in their own bands. Clock prices also require timing occupations. Concavity and common scaling apply to the expanded price vector for a fixed feasible set, but one constant A/c root need not solve the banded controller. This makes the present catalog useful for price geometry without mistaking it for a provider deployment ranking.

L Task Efficiency Beyond Conditional Invoice Minimization

L.1 The estimand and its denominator

Fix the issue population, tools, model, evaluator and stopping/retry contract. Let Y indicate verified completion and ℓ count all four billed categories over the entire attempt, including failure. With $T = \mathbf{1}^\top \ell$ and $C = p^\top \ell$, identically distributed attempts with finite means and positive resource mean satisfy, by the law of large numbers,

$$\frac{\sum_r Y_r}{\sum_r T_r} \longrightarrow \frac{\mathbb{E}Y}{\mathbb{E}T}, \quad \frac{\sum_r Y_r}{\sum_r C_r} \longrightarrow \frac{\mathbb{E}Y}{\mathbb{E}C}. \quad (84)$$

The numerator is accepted tasks, not calls or self-reported success. Repeated cached tokens count as repeated billed processing in T ; dollars weight categories differently. Unique information or generated tokens can be additional diagnostics, but changing the denominator changes the named metric. An average of individual Y/T ratios gives a different weighting, while $\mathbb{E}[T \mid Y = 1]$ omits failure expense. If independent complete attempts are repeated with a fixed policy until success, expected aggregate expense obeys

Table 19: All GPT/OpenAI configurations. H_q is the marked line in thousand tokens at controlled q . Last-column block saving compares retuning with keeping 112.848k at $q = 1$, not with another model. Labels identify catalog configurations; short/long are fixed quote projections.

Quoted configuration	USD/M (i, w, r, o)	H_8 (k)	H_{95} (k)	H_1 (k)	Block saving (%)
gpt53codex standard	(1.75, 1.75, 0.175, 14)	94.208	105.448	113.728	0.015
gpt6astra short	(10, 12.5, 1, 50)	91.003	102.938	112.848	0.000
gpt6astra long	(20, 25, 2, 75)	90.068	101.573	111.133	0.024
gpt61sol short	(2, 2.5, 0.1, 10)	92.663	110.393	132.838	1.566
gpt61sol long	(4, 5, 0.2, 15)	91.678	108.748	130.383	1.326
gpt6luna short	(0.1, 0.125, 0.01, 0.5)	91.003	102.938	112.848	0.000
gpt6luna long	(0.2, 0.25, 0.02, 0.75)	90.068	101.573	111.133	0.024
gpt6sol short	(2, 2.5, 0.2, 10)	91.003	102.938	112.848	0.000
gpt56sol short	(4, 5, 0.4, 20)	91.003	102.938	112.848	0.000
gpt56sol long	(8, 10, 0.8, 30)	90.068	101.573	111.133	0.024
gpt56terra short	(2, 2.5, 0.2, 12)	91.723	103.998	114.183	0.028
gpt56luna short	(0.2, 0.25, 0.02, 1.2)	91.723	103.998	114.183	0.028
gpt54 short	(2.5, 2.5, 0.25, 15)	92.618	103.198	110.993	0.027
gpt54 long	(5, 5, 0.5, 22.5)	91.368	101.418	108.838	0.102
gpt54mini short	(0.75, 0.75, 0.075, 4.5)	92.618	103.198	110.993	0.027
gpt52 short	(1.75, 1.75, 0.175, 14)	94.208	105.448	113.728	0.015
gpt41 short	(2, 2, 0.5, 8)	86.893	90.613	92.403	3.591

Table 20: All remaining Claude/Qwen/DeepSeek configurations with the same units and controls. SG/BJ denote Singapore/Beijing, explicit/implicit the quoted cache mode. Provider request-band switching is not simulated.

Quoted configuration	USD/M (i, w, r, o)	H_8 (k)	H_{95} (k)	H_1 (k)	Block saving (%)
sonnet46 5m	(3, 3.75, 0.3, 15)	91.003	102.938	112.848	0.000
sonnet55 5m	(2, 2.5, 0.2, 10)	91.003	102.938	112.848	0.000
opus55 5m	(4, 5, 0.2, 20)	92.663	110.393	132.838	1.566
fable51 5m	(10, 12.5, 0.25, 50)	93.653	116.288	161.263	5.470
sonnet55 1h	(2, 4, 0.2, 10)	89.543	104.888	121.578	0.430
qwen38max BJ explicit	(1.65, 2.063, 0.137, 4.951)	89.993	102.703	114.483	0.015
qwen38max BJ implicit	(1.65, 1.65, 0.206, 4.951)	89.253	97.078	102.223	0.825
qwen38max SG explicit	(2, 2.5, 0.17, 6)	89.918	102.458	113.898	0.020
qwen38max SG implicit	(2, 2, 0.25, 6)	89.248	97.063	102.198	0.828
qwen38flash BJ explicit	(0.113, 0.177, 0.014, 0.382)	88.683	100.108	109.583	0.093
qwen38flash BJ implicit	(0.113, 0.113, 0.014, 0.382)	89.608	97.608	102.888	0.711
qwen38flash SG explicit	(0.15, 0.2, 0.016, 0.47)	89.263	100.538	109.888	0.063
qwen38flash SG implicit	(0.15, 0.15, 0.016, 0.47)	89.953	98.998	105.458	0.373
qwen37plus SG short explicit	(0.4, 0.5, 0.04, 1.6)	90.248	101.853	111.478	0.013
qwen37plus SG short implicit	(0.4, 0.4, 0.08, 1.6)	88.008	92.948	95.553	2.428
qwen37plus SG long explicit	(1.2, 1.5, 0.12, 4.8)	90.248	101.853	111.478	0.013
qwen37plus SG long implicit	(1.2, 1.2, 0.24, 4.8)	88.008	92.948	95.553	2.428
deepseek41flash offpeak	(0.15, 0.15, 0.003, 0.6)	94.423	116.923	162.003	5.585
deepseek41flash peak	(0.3, 0.3, 0.006, 1.2)	94.423	116.923	162.003	5.585
deepseek4pro offpeak	(0.66, 0.66, 0.022, 1.98)	92.753	111.018	137.273	2.262
deepseek4pro peak	(1.32, 1.32, 0.044, 3.96)	92.753	111.018	137.273	2.262

Table 21: Matched- q Qwen mode comparisons after separately retuning both. Positive excess means explicit is more expensive. Block Δ is explicit minus implicit, 10^{-3} USD/action with paired MC95 halfwidths.

Qwen quoted model	q	Explicit excess (%)	Block $10^3 \Delta$
qwen38max SG	0.8	+6.865	+4.27071 $\pm$ 0.02843
qwen38max SG	0.95	-6.514	-2.68776 $\pm$ 0.02025
qwen38max SG	1	-15.973	-5.44643 $\pm$ 0.00849
qwen38flash SG	0.8	+20.533	+0.92551 $\pm$ 0.00234
qwen38flash SG	0.95	+12.385	+0.35939 $\pm$ 0.00166
qwen38flash SG	1	+6.298	+0.14621 $\pm$ 0.00076
qwen37plus SG short	0.8	-4.341	-0.63765 $\pm$ 0.00692
qwen37plus SG short	0.95	-21.581	-2.36563 $\pm$ 0.00464
qwen37plus SG short	1	-31.375	-3.03948 $\pm$ 0.00209

$R = \mathbb{E}C + (1-s)R$, hence $R = \mathbb{E}C/s$. Within-attempt cost and success may be dependent. Adaptive retry behavior after failure requires a different joint policy, not this fixed-attempt recurrence.

L.2 Cost ranking and efficiency sensitivity

For positive expected resource $r(H)$ and success $s(H)$,

$$\eta' = \frac{s'r - sr'}{r^2}, \quad \eta' = 0 \iff s'/s = r'/r \quad (85)$$

at a positive-success smooth interior optimum. A cost-only point with $C' = 0$ has dollar-efficiency slope s'/C ; it is stationary only if $s' = 0$. Cross multiplication of two positive ratios proves the success-loss boundary in Discussion. A cheaper policy can lose useful throughput without contradicting any invoice calculation.

Assume policy-indexed $s(H)$ and $\mathbf{m}(H)$ are explicitly independent of prices. For $C(H, \mathbf{p}) = \mathbf{p}^\top \mathbf{m}(H)$, set $F = s'C - sC'$. At a nondegenerate active efficiency root, $F_H = s''C - sC''$ and $F_{p_j} = s'm_j - sm'_j$. Thus

$$\partial_{p_j} H_* = -\frac{s'm_j - sm'_j}{s''C - sC''}. \quad (86)$$

Success gradients and curvature enter both numerator and denominator. Constant success reduces this to $-m'_j/C''$, the ordinary cost-optimum response. Common positive price scaling leaves efficiency policy choices unchanged while dividing dollar efficiency by that scale. Changing model behavior or price-dependent budget feasibility changes the kernel/policy problem rather than this partial derivative. The companion SymPy derivation checks quotient, stationary, price-response and scaling identities.

L.3 Fractional optimization and a success constraint

For a finite policy family with positive C_π , define $\lambda_* = \max_\pi s_\pi/C_\pi$. Then $s_\pi - \lambda_* C_\pi \leq 0$ for every policy, with equality at an optimizer. Conversely, an attained zero maximum of that difference identifies a ratio optimizer. General families require supremum and attainment/boundedness conditions. This is the standard fractional-programming reduction [25].

In a finite observed-state task model, terminal reward is verified success and stage reward is $-\lambda \mathbf{p}^\top \mathbf{m}_n(s, a)$:

$$W_n(s) = \max_a \{-\lambda \mathbf{p}^\top \mathbf{m}_n(s, a) + \sum_{s'} P_n(s'|s, a) W_{n+1}(s')\}. \quad (87)$$

Success/failure terminal values replace an ordinary-action counter as the achievement criterion. The kernel must capture lost obligations, retrieval, retries and changed future work; an independent success penalty attached to the old fixed-growth curve does not generally do so.

Alternatively minimize C_π subject to $s_\pi \geq \tau$. When task law is fixed, this feasible set is price-independent, so optimized invoice concavity, monotonicity and homogeneity still follow from its lower envelope. The currently unconstrained cost optimum need not satisfy the floor. The success-expense frontier makes that distinction explicit. This preserves the useful invoice geometry without elevating it to a complete task-performance model.

L.4 A discriminating task-level experiment

Intervene on compact controllers over a fixed issue cohort with an external evaluator. Stratify by cross-file obligations and task horizon; hold model, tools, initial repository snapshot, per-attempt limits and retry protocol fixed. Compare the broad conditional-cost basin with an earlier line, a later line and a state-aware restoration policy. Pair task instances across policies to control difficulty; replicate stochastic attempts independently. Future calls and token counts must be allowed to change, because that is part of the policy effect.

Record verified success, all billed categories, recovery, latency, retries, stopping reasons, retained obligations and failure modes. Count every assigned instance, including exhausted budgets and failures. Report success rate, pooled tasks per billed token, tasks per dollar and the cost-success frontier. Bootstrap task instances as clusters, retaining their paired policy outputs; thousands of calls from one issue are not independent task evidence. A useful intervention preserves the declared success floor or improves throughput, not just its fixed-call invoice. The current usage-only pool has no paired policy-intervention success labels. It measures the resource mechanism; the missing success frontier is the next consequential empirical measurement, not a quality curve to fabricate from usage counters.

M Execution, Verification, and Artifact Mapping

M.1 Deterministic renewal calculation

Each empirical increment is rounded to the nearest multiple of five, with maximum perturbation 2.5 tokens. The renewal mass satisfies

$$r_0 = \frac{1}{1 - f_0}, \quad r_k = \frac{\sum_{i=1}^k f_i r_{k-i}}{1 - f_0}, \quad (88)$$

where f_i is binned probability mass. Prefix sums supply U , M , and D ; crossing-increment contributions supply z . Zero growth remains in f_0 . The evaluated gap range is 250k tokens, with an unbounded-right-tail certificate: beyond the bracketed core crossing the core rate is nondecreasing, and the finite support bounds any negative terminal term by $d\theta G_{\max}/U_{\text{end}}$. The lower bound exceeds each selected minimum in the reported cases.

An independent category-level residual-state dynamic program reconciles Equation (35) on 45 constructed-law cases to a maximum 8.53×10^{-14} tokens/action. Finite contrasts match nonatom price derivatives to 2.44×10^{-10} relative error and the smooth price Hessian to 3.32×10^{-7} . SymPy checks the exact sign factors, marked-tail derivatives, material paths, homogeneity and regret identities. Lean checks finite invoice and renewal comparison laws, including the repriced-resource ordering, using the installed Lean 4.33.1 kernel.

M.2 Sampling and uncertainty

The original community study uses 512 discovery trajectories per dependence mode with seeds 2026100601 and 2026100602, and 512 independent confirmation trajectories with seeds 2026101601 and 2026101602. Horizons are 32, 400 and 4,000 ordinary actions. The tariff study uses seeds 2026102601 and 2026102602, 512 replicates, and 4,000 actions. Shared growth/output/cache fields pair thresholds and mechanism contrasts. Price-only groups use one immutable array of the four token categories and dot it with each price vector.

For paired per-action differences Δ_r , the reported interval is $\bar{\Delta} \pm 1.96s_{\Delta}/\sqrt{512}$. Cost-rate variance comparisons use the same ordinary-action horizon and matched marginal cohort. The long-horizon 1% basin is a descriptive region of sampled mean invoices, rather than a confidence set for a population optimum. Exploratory choices are kept distinct from independent confirmation.

M.3 Reusable artifacts

The public implementation and numerical artifacts are available at <https://github.com/kleedaisuki/context-compaction-lab>. Portable result files reside in docs/research. The expanded tariff study retains all 35 earlier analytical scenarios exactly. Its 143 cases include 38 vectors at three q values and 29 retained mechanism controls. Figures in this paper are generated from those artifacts and the pinned empirical pool, with embedded plot coordinates for standalone compilation.

```
uv sync --locked
uv run python experiments/calibrate_renewal_trace.py
uv run python experiments/renewal_community_study.py
uv run python experiments/run_price_sensitivity.py
    --cache-conditions 0.8 0.95 1
    --result-name price-family-results
uv run python experiments/price_family_crosscheck.py
uv run python experiments/build_paper_figures.py
uv run python experiments/build_paper_exhibits.py
```

The task-efficiency symbolic runner is listed in Table 3. The multi-line tariff command is shown wrapped for readability and is run as one shell command. The trace must first be downloaded using the pinned release URL and checksum documented by the calibration script. Generated raw pools, intermediate logs and compiler objects stay under the repository cache. The exhibits use maintained aggregates or the exact exported jump law; regenerating presentation makes no new model call or sampled trajectory.